Adaptive quantum tomography by least squares with optimal asymptotic covariance

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Abstract: Adaptive quantum state tomography, which uses previous measurement outcomes to select subsequent measurements, enables more efficient and accurate state estimation in quantum information processing. We study explicit least-squares procedures for interior mixed qubits and pure states in every fixed finite dimension, with measurements updated between batches. We align fixed reference measurements with current estimates, update measurement selection probabilities under the mixed-qubit Frobenius optimal design, and give a pure-state implementation using a Householder unitary followed by randomized projective measurement. Safeguarded ridge least squares for mixed qubits uses all observations to update the design and obtain the estimate. Corrected least squares for pure states uses a preliminary ridge fit to align a Fisher-symmetric reference measurement and a weighted correction on the fitted tangent space to obtain the estimate. We prove strong consistency, convergence of the conditional classical Fisher information to loss-optimal targets, and asymptotic normality with inverse-information covariance. The correction can be essential: the preliminary fit can retain excess asymptotic variance even when measurement information converges to its optimum. Under the design matched to each loss, the reported estimators attain the Gill–Massar benchmark at each fixed true state in asymptotic expected squared Bures or Frobenius loss.

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1. Introduction

Quantum information processing requires reliable descriptions of the states prepared and manipulated by an experiment. Quantum state tomography estimates such a state from measurements on independently prepared copies of a system (Nielsen and Chuang, 2010). The choice of measurement affects precision, and a good choice generally depends on the unknown state. Adaptive tomography addresses this difficulty through a feedback loop: collect outcomes, update a state estimate, and use it to select measurements for the next copies. Photonic experiments have demonstrated gains from adaptive qubit tomography and the feasibility of repeated measurement updates (Mahler et al., 2013; Hou et al., 2016; Okamoto et al., 2017).

Statistically, this feedback loop combines sequential experimental design with estimation from conditionally categorical observations (Kiefer, 1959; Chernoff, 1959; Chaudhuri and Mykland, 1993; Li and Zhao, 2025). Fisher information measures how sensitively the outcome probabilities change near the true state. A loss specifies which estimation errors matter, so the desired information depends on the loss. In the single-copy models studied here, the

Gill–Massar inequality limits the information available from any measurement (Gill and Massar, 2000). Three questions then arise: does the adaptive design approach the loss-optimal information, does the estimator use that information with inverse-information covariance, and does its expected physical loss attain the corresponding asymptotic constant? These are distinct claims; information convergence alone does not answer the other two.

Least squares is natural because measurement probabilities are linear in the density matrix (Qi et al., 2013). But categorical observations have unequal variances, and a good measurement design need not make an unweighted fit efficient. Our pure-state example makes this distinction explicit: the measurement information converges to its optimal target, while the preliminary least-squares fit retains excess asymptotic variance (4.7). We analyze explicit procedures that attain both the information target and inverse-information covariance, and establish their pointwise expected-loss limits. The models are interior mixed qubits and pure states in every fixed finite dimension.

For mixed qubits, a safeguarded ridge least-squares fit to all previous observations provides both the reported estimate and the estimate used to choose subsequent measurements. Ridge regularization makes the fit unique even before the measured directions span the parameter space; it does not by itself guarantee asymptotic efficiency. We use the established Bures and Frobenius allocations (Gill and Massar, 2000; Hou et al., 2016) and prove that repeated updates yield strong consistency, convergence to the corresponding information targets, and asymptotic normality with inverse-information covariance (Theorems 3.1–3.2). The covariance argument also covers the maximally mixed state, where no radial direction is distinguished.

For pure states, the preliminary and reported estimates have different roles. A ridge matrix fit is projected onto rank-one projectors and selects the next measurement alignment. The reported estimator adds a weighted least-squares correction in the fitted tangent space, which describes the possible first-order changes of a pure state, and then normalizes back to the pure-state model. The weights account for the outcome probabilities. A phase-paired reference measurement combines global identification with Fisher symmetry, a local precision property (Proposition 4.1). Informational completeness identifies the state, while Fisher symmetry gives classical information equal to one half of the quantum Fisher information at the reference state. Adaptive alignment moves this local precision to the estimated state. Theorems 4.2 and 4.3 establish consistency, information convergence and the corrected estimator's Gaussian limit.

The measurement construction separates fixed reference apparatus from adaptive effective measurements. For mixed qubits, the estimate selects three orthogonal Bloch axes; the Frobenius rule also adapts their sampling probabilities. The real Householder matrix constructs these axes. For pure states, a phase-adjusted Householder unitary aligns a fixed rank-one reference POVM with the preliminary estimate. Example 4.1 realizes the reference POVM by randomization over fixed projective bases. Applying the adjoint alignment unitary before this readout changes the effective measurement while keeping those reference bases fixed. The pure-state results also allow reference POVMs without this particular projective implementation.

The design and estimation ingredients have close precedents. Gill and Massar (2000, Theorem VII) prove a Gaussian limit and a mean-square-error expansion for two-stage adaptive qubit estimation; Fujiwara (2006, 2011) develops consistency and efficiency theory for adaptive likelihood methods. Linear regression tomography has been combined with adaptive measurement selection (Qi et al., 2017) and ridge regularization (Mu et al., 2020). For fixed designs, Acharya, Kypraios and Guță (2019) analyze generalized least squares with fitted

probability weights. In pure-state estimation, Li et al. (2016) obtain Fisher symmetry from two phase-related projective bases, whereas no fixed finite-outcome single-copy measurement has this property at every pure state (Zhu and Hayashi, 2018). Girotti, Godley and Guţă (2024) establish Gaussian limits and asymptotic expected-risk attainment for displaced-null measurements, and Vargas, Delgado and Pereira (2026) study a three-stage Fisher-symmetric protocol with reconstruction bounds and likelihood refinement. Our contribution is the joint information, covariance and expected-loss analysis of the specified repeatedly updated least-squares procedures, including the fitted tangent correction.

The main proof difficulty is reuse of adaptive observations: they guide later measurements and also determine the tangent space and probability weights used in the final correction. Expanding around martingale sums at the true state controls these fitted quantities and removes the preliminary error to first order. This uses stochastic regression and adaptive-design arguments (Lai and Wei, 1982; Pronzato, 2010; Freise, Gaffke and Schwabe, 2021) together with one-step estimation (Brouste, Dutang and Noutsa Mieniedou, 2021). Measurement information can converge even when the full alignment unitary does not. Updates may follow any predetermined sequence of positive finite batch sizes, including single-copy updates; estimator limits hold at completed batches. Optimality is pointwise at each fixed truth and dimension. The finite-sample comparison in Theorem 5.1 concerns locally unbiased coordinate estimators. The reported physical estimates may be biased; their asymptotic expected-loss attainment follows from the Gaussian limits and proved moment bounds (Corollaries 5.2–5.3). The loss relations in Section 2.5 also give infidelity and squared full trace-norm constants. We do not assert uniform local minimax attainment or finite-sample superiority to likelihood estimation.

Section 2 introduces the models and information benchmarks. Sections 3 and 4 present the mixed-qubit and pure-state procedures. Section 5 gives the adaptive bound and expected-loss attainment, and Section 6 illustrates information and physical risk. Appendices A and B provide the geometry and conditional-likelihood framework. Each theoretical result is followed by its proof location.

2. Statistical model and information benchmarks

2.1. Quantum states, measurements and adaptive design

We consider a D-level quantum system: its Hilbert space is $\mathbb{C}^D$, with D orthonormal basis states (Bertlmann and Krammer, 2008, Section 2). A qubit is a two-level system. For a system of q qubits, the Hilbert-space dimension is $D = 2^q$. We observe independently prepared copies of a common unknown $D \times D$ density matrix ρ_0, a complex Hermitian positive semidefinite matrix satisfying $\mathrm{tr}(\rho_0) = 1$, with $D \geq 2$. For interior mixed qubits, ρ_0 is positive definite; for pure states, it has rank one (Nielsen and Chuang, 2010, Sections 1.2, 2.2.8 and 2.4.2). We study interior mixed qubits ($D = 2$) and pure states of D-level systems. Our goal is loss-optimal adaptive tomography for the four losses defined in Section 2.5.

Write $\mathcal{A}$ for the action space, $a \in \mathcal{A}$ for a fixed possible action, and $a_t \in \mathcal{A}$ for the action selected on copy t. The time subscript indexes a selected value of the same object. We use two vector action spaces. The mixed-qubit measurements are projective; the pure-state measurements are finite rank-one POVMs, with a randomized-projective implementation in Example 4.1. For mixed qubits, $\mathcal{A} = \{a \in \mathbb{R}^3 : \|a\| = 1\}$; the action selects the Pauli

observable $a \cdot \sigma$ described in Section 2.2. For pure states, $\mathcal{A} = \{a \in \mathbb{C}^D : \|a\| = 1\}$: an action specifies the measurement alignment, and the fixed Householder map in (4.6) in Section 4.4 constructs the corresponding POVM from the vector a. The model-specific measurements and sampling laws are given in Sections 3.1 and 4.4. For Hermitian matrices A, B of the same size, $A \preceq B$ (or $B \succeq A$) means that $B-A$ is positive semidefinite. The strict relation $A \prec B$ means that $B - A$ is positive definite. In our statistical model, a *measurement setting* is the measurement specified by an action a, represented by a finite positive operator-valued measure (POVM), a collection of effects $E_{a,y} \succeq 0$ satisfying $\sum_y E_{a,y} = \mathrm{I}_D$. A projection-valued measure (PVM), or projective measurement, is a POVM whose effects are mutually orthogonal projections (Nielsen and Chuang, 2010, Sections 2.2.5–2.2.6). Randomizing among finitely many PVMs and recording both the choice and outcome gives a POVM with probability-weighted projection effects. Here I_m is the $m \times m$ identity matrix. The realized effect $E_{a_t,y}$ is this same family evaluated at the selected action a_t. An effect is the POVM element for one outcome label y; it satisfies $0 \preceq E_{a,y} \preceq \mathrm{I}_D$ in this matrix order. For a complex matrix or column vector, $A^* = \overline{A}^{\mathsf{T}}$ denotes its conjugate transpose (adjoint), while A^{T} denotes ordinary transpose. A rank-one projective measurement has effects $b_k b_k^*$, where the vectors b_k form an orthonormal basis of $\mathbb{C}^D$.

We now describe the adaptive design, the selection of measurement actions, and the resulting observation history. On each copy, the available history determines the sampling law for the next action; the selected action specifies the measurement, and the new record (O_t, a_t) extends the history. Let $\mathcal{F}_0$ be the sigma-field of information available before the first measurement, trivial for deterministic initialization. For each $t \geq 1$, choose a random design ξ_t, an $\mathcal{F}_{t-1}$-measurable probability measure on the action space $\mathcal{A}$. The notation $a_t \sim \xi_t$ always refers to the conditional law given the past:

$$\mathbb{P}\{a_t \in C \mid \mathcal{F}_{t-1}\} = \xi_t(C) \quad \text{for every measurable set of actions } C.$$

Draw the action from this law and record O_t with conditional Born probabilities $\mathrm{tr}(\rho_0 E_{a_t,y})$ (Nielsen and Chuang, 2010, Section 2.4.1). The design may be a point mass, in which case the past fixes the action. If ξ_t is discrete, write $\xi_t(a) = \xi_t(\{a\})$ for its mass at action a; the joint conditional distribution is

$$\mathbb{P}(a_t = a, O_t = y \mid \mathcal{F}_{t-1}) = \xi_t(a)\,\mathrm{tr}(\rho_0 E_{a,y}).$$

After this record, set $\mathcal{F}_t = \sigma(\mathcal{F}_{t-1}, a_t, O_t)$. Let $\mathbb{E}$ denote expectation.

We use O_t for a generic outcome, $Y_t \in \{-1, 1\}$ for a mixed-qubit response, and Z_t for a pure-state categorical outcome. Section 4.2 defines the complete pure-state label space $\mathcal{Z}$.

In this single-copy model, one physical shot prepares a fresh copy of ρ_0 and measures it once, producing one setting–outcome record. Fix predetermined positive finite batch sizes $m_1, m_2, \ldots$ and let m_j count the shots in batch j. Define

$$N_0 = 0, \qquad N_j = \sum_{k=1}^{j} m_k, \qquad n_-(t) = N_{j-1} \quad (N_{j-1} < t \leq N_j). \tag{2.1}$$

Thus N_j is the cumulative shot count through batch j. For batch j, with $N_{j-1} < t \leq N_j$, the measurement design is determined using only the tomography estimate based on the first N_{j-1} observations and is held fixed throughout that batch. Sections 3.1 and 4.4 give the resulting mixed-qubit and pure-state measurement rules.

Let $\mathrm{Herm}(\mathbb{C}^D)$ denote the real vector space of $D \times D$ Hermitian matrices. On this space write $\langle A, B\rangle_F = \mathrm{tr}(AB)$. The symbols $\|\cdot\|$, $\|\cdot\|_F$, $\|\cdot\|_1$ and $\|\cdot\|_{\mathrm{op}}$ denote Euclidean, Frobenius, trace and induced operator norms. Unless specified otherwise, Hermitian errors and operators on them use the Frobenius norm and its induced norm. A subscript zero denotes the truth. We use $\Longrightarrow$ for convergence in distribution, $\xrightarrow{\mathbb{P}}$ for convergence in probability, and a.s. for almost surely. The symbols O_p and o_p have their usual stochastic order meanings; $a_n \gg b_n$ means $b_n/a_n \to 0$.

2.2. *Mixed qubits: density matrices and information*

The three Pauli matrices form a real basis for traceless Hermitian 2×2 matrices (Bertlmann and Krammer, 2008, Section 3.1 and Eq. (35)). They give the Bloch coordinates of every qubit state (Zhong et al., 2013, Eq. (10)):

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho(\theta) = \frac{\mathrm{I}_2 + \sum_{k=1}^3 \theta_k \sigma_k}{2}. \tag{2.2}$$

Here $\theta \in \mathbb{R}^3$ is the Bloch vector. The eigenvalues of $\rho(\theta)$ are $(1 \pm \|\theta\|)/2$, so the state space is the closed unit ball. The interior $\|\theta\| < 1$ comprises the full-rank mixed qubits studied in Section 3; its boundary comprises pure qubits.

For a unit vector $a \in \mathbb{R}^3$, the observable $a \cdot \sigma = \sum_{r=1}^3 a_r \sigma_r$ satisfies $(a \cdot \sigma)^2 = \mathrm{I}_2$. Measuring it is therefore the two-outcome projective measurement with effects $E_{a,y} = (\mathrm{I}_2 + y\, a \cdot \sigma)/2$, $y \in \{-1, 1\}$. The Born rule and its derivative give the outcome probabilities and classical Fisher information matrix, respectively (Barndorff-Nielsen and Gill, 2000, Sections 2 and 3):

$$\begin{aligned} p_{a,y}(\theta) &= \mathrm{tr}\{\rho(\theta) E_{a,y}\} = \frac{1 + y\, a^{\mathsf{T}}\theta}{2}, \\ \mathcal{I}_a(\theta) &= \sum_{y=\pm 1} \frac{\nabla p_{a,y}(\theta) \nabla p_{a,y}(\theta)^{\mathsf{T}}}{p_{a,y}(\theta)} = \frac{a a^{\mathsf{T}}}{1 - (a^{\mathsf{T}}\theta)^2}. \end{aligned} \tag{2.3}$$

Thus the action a determines the two projectors $E_{a,\pm 1}$; for a fixed state, its role in the Born law is through the scalar $a^{\mathsf{T}}\theta$. For example, $a = (1, 0, 0)^{\mathsf{T}}$ measures σ_1 and gives probabilities $(1 \pm \theta_1)/2$. On copy t, substitute the selected action a_t into the same effect family. At the true state $\rho_0 = \rho(\theta_0)$,

$$\mathbb{P}\{Y_t = y \mid \mathcal{F}_{t-1}, a_t\} = \mathrm{tr}\{\rho(\theta_0) E_{a_t,y}\} = \frac{1 + y\, a_t^{\mathsf{T}}\theta_0}{2}, \qquad y \in \{-1, 1\}.$$

The design ξ_t selects the axis; this conditional Born law gives the outcome distribution after the selection. In particular, $\mathbb{E}(Y_t \mid \mathcal{F}_{t-1}, a_t) = a_t^{\mathsf{T}}\theta_0$, which is the regression relation used in Section 3.1. The Fisher matrix $\mathcal{I}_a(\theta)$ has rank one: this measurement is informative only along a. For $\|\theta\| < 1$, the quantum Fisher information matrix in these coordinates, based on symmetric logarithmic derivatives (SLDs), is (Gill and Massar, 2000, Eq. (8))

$$H(\theta) = \mathrm{I}_3 + \frac{\theta\theta^{\mathsf{T}}}{1 - \|\theta\|^2}, \qquad H(\theta)^{-1} = \mathrm{I}_3 - \theta\theta^{\mathsf{T}}.$$

The general definition of SLDs and the coordinate interpretation of quantum Fisher information are given in Section 2.4.

2.3. Pure states and tangent directions

Now let $D \geq 2$. A state on $\mathbb{C}^D$ is a $D \times D$ density matrix, and it is pure exactly when $P = hh^*$ for a unit vector $h \in \mathbb{C}^D$. For the pure-state model, write $P_0 = h_0 h_0^*$ for the unknown true state. In Dirac notation, the ket $|h\rangle$ represents the vector h, the bra $\langle h|$ its adjoint h^*, and $|h\rangle\langle h| = hh^*$. We use column-vector and matrix notation. The subscript ref labels a fixed reference object, while 0 denotes the true state or parameter. A superscript $\star$ labels an oracle optimum, such as $\mathcal{I}^\star$, rather than an adjoint. The mixed-qubit model uses three real parameters; the pure model below uses $2(D-1)$ and includes pure qubits as $D = 2$. The pure-state manifold is $\mathcal{P}_D = \{P \in \mathrm{Herm}(\mathbb{C}^D) : P^2 = P,\ \mathrm{tr}\, P = 1\}$. It has real dimension $2d$, where $d = D - 1$. Projectors identify unit vectors differing only by a global phase. Appendix A, Definition A.1, describes the corresponding projective space and its identification with $\mathcal{P}_D$. At $P = hh^*$, define the complex orthogonal complement $h^\perp = \{v \in \mathbb{C}^D : h^* v = 0\}$. It has complex dimension $D-1$ and real dimension $2(D-1)$; the vectors in this space have arbitrary norm. The tangent space (Definition A.3) is

$$T_P = \{A = A^* : PA + AP = A\} = \{hv^* + vh^* : v \in h^\perp\}.$$

A tangent vector describes the first-order change along a curve of pure states.

Section 4.1 gives local coordinates when they are needed to analyze projection and the tangent correction.

2.4. Classical and quantum Fisher information

For a finite POVM $\mathsf{M} = \{M_y\}$ and a pure state P, put $p_y(P) = \mathrm{tr}(PM_y)$. Its classical Fisher information is the coordinate-free bilinear form $\mathcal{J}_{\mathsf{M},P} : T_P \times T_P \longrightarrow \mathbb{R}$, given by the expected product of directional scores (Barndorff-Nielsen and Gill, 2000, Section 2):

$$\mathcal{J}_{\mathsf{M},P}[A, B] = \sum_{y : p_y(P) > 0} \frac{\mathrm{tr}(M_y A)\,\mathrm{tr}(M_y B)}{p_y(P)}, \qquad A, B \in T_P.$$

This definition requires no coordinates. A positive definite form identifies every tangent direction locally. It varies smoothly where all outcome probabilities are positive. At a zero probability we use the displayed positive-support convention, without assuming continuity there. Reparametrization cannot repair a singular experiment.

We write Γ for a parametrization map and P for an individual projector. Choose a pure-state parametrization $\Gamma(\theta)$ as in Definition A.2, with $\theta \in \mathbb{R}^p$ and $p = 2(D-1)$. Section 4.1 gives the explicit parametrization (4.2). At $P = \Gamma(\theta)$, let $A_j = \partial_j \Gamma(\theta)$ be its coordinate tangent basis. Its matrix in these coordinates is

$$[\mathcal{I}_{\mathsf{M}}(\theta)]_{jk} = \mathcal{J}_{\mathsf{M},\Gamma(\theta)}[A_j, A_k], \qquad j, k = 1, \ldots, p. \tag{2.4}$$

For simplicity, the notation $\mathcal{I}_{\mathsf{M}}(\theta)$ omits the chosen parametrization $\Gamma(\theta)$; the matrix depends on that choice. For an action a selecting $\mathsf{E}_a = \{E_{a,y}\}$, we write $\mathcal{J}_{a,P} := \mathcal{J}_{\mathsf{E}_a,P}$ and $\mathcal{I}_a(\theta) := \mathcal{I}_{\mathsf{E}_a}(\theta)$. Thus $\mathcal{J}_{a,P}$ acts on tangent vectors, whereas $\mathcal{I}_a(\theta)$ is a $p \times p$ matrix acting on coordinate vectors. If $A = \sum_j u_j A_j$ and $B = \sum_k w_k A_k$, then

$$\mathcal{J}_{a,\Gamma(\theta)}[A, B] = u^{\mathsf{T}} \mathcal{I}_a(\theta) w.$$

Here $p_{a,y}(\theta) = \mathrm{tr}\{E_{a,y}\Gamma(\theta)\}$, so $\partial_j p_{a,y}(\theta) = \mathrm{tr}(E_{a,y}A_j)$; substituting into (2.4) gives the usual coordinate Fisher matrix, with the same positive-support convention. For interior mixed qubits, the same score-based definition applied to the Born probabilities in (2.3) gives the matrix in that equation (Gill and Massar, 2000, Eq. (20)). Here $p = 3$ for mixed qubits and $p = 2(D-1)$ for pure states. In what follows, $\rho = \rho(\theta)$ and $\partial_j\rho = \partial\rho(\theta)/\partial\theta_j$; all derivatives and SLDs are evaluated at the same coordinate value θ.

Quantum Fisher information matrices have size $p \times p$, whereas effects and states have size $D \times D$. Given local coordinates $\theta_1, \ldots, \theta_p$, first define the Hermitian SLDs Λ_j by solving

$$\partial_j\rho = \frac{\rho\Lambda_j + \Lambda_j\rho}{2}, \qquad j = 1, \ldots, p.$$

For interior mixed qubits, $\rho \succ 0$ makes the solution unique: in an eigenbasis of ρ with eigenvalues $\lambda_r > 0$, $(\Lambda_j)_{rs} = 2(\partial_j\rho)_{rs}/(\lambda_r + \lambda_s)$ (Gill and Massar, 2000, Eq. (22)). For pure states $\rho = \Gamma(\theta)$, one may take $\Lambda_j = 2A_j$ (Facchi et al., 2010, Eqs. (17)–(18)). Define the SLD quantum Fisher information matrix (Gill and Massar, 2000, Eqs. (24)–(25)) by

$$[H(\theta)]_{jk} = \frac{1}{2}\,\mathrm{tr}\{\rho(\Lambda_j\Lambda_k + \Lambda_k\Lambda_j)\}, \qquad j, k = 1, \ldots, p.$$

For pure states, any other SLD differs by a Hermitian matrix K_j with $PK_j = K_jP = 0$; these changes leave $H(\theta)$ unchanged (Liu et al., 2014, Eqs. (5)–(7)). Thus $H(\theta)$ is well defined for both models.

For pure states, directional SLDs Λ_A and Λ_B are Hermitian solutions of

$$A = \frac{P\Lambda_A + \Lambda_AP}{2}, \qquad B = \frac{P\Lambda_B + \Lambda_BP}{2}.$$

The coordinate-free SLD quantum Fisher information form $\mathcal{H}_P : T_P \times T_P \longrightarrow \mathbb{R}$ is

$$\mathcal{H}_P[A, B] = \frac{1}{2}\,\mathrm{tr}\{P(\Lambda_A\Lambda_B + \Lambda_B\Lambda_A)\}.$$

The next proposition gives a formula for the coordinate-free quantum Fisher information form that does not depend on the choice of SLDs.

Proposition 2.1 (Pure-state quantum Fisher information)**.** *Let $P = hh^*$ with $\|h\| = 1$. The pairing $\mathcal{H}_P$ is independent of the chosen directional SLDs. For $A, B \in T_P$ and $v \in \mathbb{C}^D$ with $h^*v = 0$, it satisfies*

$$\mathcal{H}_P[A, B] = 2\,\mathrm{tr}(AB), \qquad \mathcal{H}_P[hv^* + vh^*, hv^* + vh^*] = 4\,\|v\|^2. \tag{2.5}$$

In a chart (Definition A.2) with tangent basis $A_j = \partial_j\Gamma(\theta)$, $[H(\theta)]_{jk} = \mathcal{H}_P[A_j, A_k]$ and $\mathcal{H}_P[A, B] = u^{\mathsf{T}}H(\theta)w$ whenever $A = \sum_{j=1}^p u_jA_j$ and $B = \sum_{k=1}^p w_kA_k$, with $u, w \in \mathbb{R}^p$.

Proof of Proposition 2.1. See Supplementary Material, Section S2.1. □

The form $\mathcal{H}_P$ depends on the model and state, not the measurement or loss; $H(\theta)$ is its coordinate matrix.

$\mathcal{H}_P$ is four times the Fubini–Study metric (Definition A.4), the standard Riemannian metric on the complex projective manifold $\mathbb{CP}^{D-1}$, under the convention $g_{\mathrm{FS}}(hv^* + vh^*, hv^* + vh^*) = \|v\|^2$ (Braunstein and Caves, 1994; Facchi et al., 2010). The coordinate matrix of $\mathcal{H}_P$ is I_{2d} at the center of the amplitude chart in Section 4.1, and $2\mathrm{I}_{2d}$ in Frobenius-orthonormal tangent coordinates.

In matrix formulas, H and $\mathcal{I}_a$ abbreviate $H(\theta)$ and $\mathcal{I}_a(\theta)$ in the stated parametrization; $\mathcal{I}$ denotes the information of a fixed or randomized design. Convergence of forms at a fixed state means norm convergence on its tangent space, equivalently matrix convergence in any fixed coordinates. Let $\theta = \theta(\eta)$ be a smooth invertible local coordinate change, with Jacobian $J = \partial\theta/\partial\eta$ at the truth. Definition A.3 explains this Jacobian for pure-state coordinates. Both coordinate vectors have dimension p. The chain rule gives $\widetilde{H} = J^{\mathsf{T}} H J$ and $\widetilde{\mathcal{I}} = J^{\mathsf{T}} \mathcal{I} J$. Hence

$$\widetilde{H}^{-1}\widetilde{\mathcal{I}} = J^{-1}H^{-1}\mathcal{I}J, \qquad \operatorname{tr}(\widetilde{H}^{-1}\widetilde{\mathcal{I}}) = \operatorname{tr}(H^{-1}\mathcal{I}). \tag{2.6}$$

A physical quadratic loss transforms as $\widetilde{W} = J^{\mathsf{T}} W J$, so $\operatorname{tr}(W\mathcal{I}^{-1})$ is likewise invariant when the information is nonsingular.

Simultaneously rotating the state, tangent directions and measurement setting preserves both coordinate-free Fisher information forms. For pure states, this permits calculations at the reference state $P_{\mathrm{ref}} = e_0 e_0^*$, the center of the chart (4.1) in Section 4.1; the adaptive procedure must still learn how to align the measurement with the unknown state.

In the adaptive experiment, derivatives are taken with the realized history and selected measurement setting fixed. For any coordinate parameter θ, the classical Fisher information matrix of the randomized design ξ_t is $\mathcal{I}_{\xi_t}(\theta) = \int \mathcal{I}_a(\theta)\, \xi_t(da)$. Its coordinate-free counterpart is $\mathcal{J}_{\xi_t,P} = \int \mathcal{J}_{a,P}\, \xi_t(da)$, with the same form-to-matrix relation (2.4). For a discrete design, this is $\mathcal{I}_{\xi_t}(\theta) = \sum_a \xi_t(a)\mathcal{I}_a(\theta)$. Thus $\mathcal{I}_{\xi_t}(\theta_0)$ denotes the conditional Fisher information matrix for the next copy at the true parameter θ_0, given $\mathcal{F}_{t-1}$. It differs from the observed information, the negative Hessian of a realized log likelihood (Lane, 2020). We reserve Σ for the covariance in a limiting Gaussian estimation law.

2.5. *Losses and the Gill–Massar inequality*

For two $D \times D$ density matrices ρ and σ, our fidelity and squared Bures conventions are (Zhu and Hayashi, 2018, Supplemental Material, Section II B)

$$F(\rho,\sigma) = \left[\operatorname{tr}\{(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\}\right]^2, \qquad D_B(\rho,\sigma)^2 = 2\{1 - F(\rho,\sigma)^{1/2}\}. \tag{2.7}$$

Infidelity is one minus fidelity; squared Bures distance is twice the difference between one and the square root of fidelity. Squared Frobenius distance sums the squared absolute entrywise errors between the two density matrices. Squared full trace-norm distance squares the sum of the singular values of their difference. Table 1 uses this full trace norm. Squared trace distance, with the conventional one-half factor in the norm, is one quarter of this loss. For a local loss expansion $L = \delta^{\mathsf{T}} W \delta + o(\|\delta\|^2)$, δ is the coordinate error and W its $p \times p$ matrix of quadratic loss weights. In the same coordinates at the truth, let $\mathcal{I}$ be the classical Fisher information matrix of a single-copy POVM and H the model's SLD quantum Fisher information matrix. The loss determines W; for nonsingular $\mathcal{I}$, the corresponding weighted A-optimality criterion is $\operatorname{tr}(W\mathcal{I}^{-1})$ (Stallings and Morgan, 2015; Li and Zhao, 2025). For the models considered here, the information $\mathcal{I}$ of any finite single-copy POVM satisfies the Gill–Massar inequality (Gill and Massar, 2000)

$$\operatorname{tr}(H^{-1}\mathcal{I}) \le D - 1. \tag{2.8}$$

For $W \succeq 0$ and positive definite $\mathcal{I}$, this implies (Zhu and Hayashi, 2018, Supplemental Material, Section II B)

$$\operatorname{tr}(W\mathcal{I}^{-1}) \ge C_{\mathrm{GM}}(W,H), \qquad C_{\mathrm{GM}}(W,H) = \frac{[\operatorname{tr}\{(H^{-1/2}WH^{-1/2})^{1/2}\}]^2}{D-1}. \tag{2.9}$$

Proof of (2.9). See Supplementary Material, Section S2.3. □

Attainment requires both a measurement design realizing an optimal information matrix $\mathcal{I}^\star$ and an estimator whose limiting covariance equals $(\mathcal{I}^\star)^{-1}$. The procedures below establish these properties for the stated losses.

The Holevo Cramér–Rao bound is a quantum lower bound on weighted mean-square error for locally unbiased multiparameter estimation, accounting for incompatibility between measurements optimal for different parameters. Under regularity conditions it is asymptotically attainable with collective measurements, which act jointly on several copies; the Gill–Massar bound above concerns single-copy measurements. For pure-state models, adaptive single-copy measurements can also attain the Holevo bound (Girotti, Godley and Guţă, 2024).

Table 1 uses H at the truth, $s_0 = (1 - \|\theta_0\|^2)^{1/2}$ for mixed qubits, and $d = D - 1$ for pure states. Constants follow from (2.9).

TABLE 1
Loss weights, Gill–Massar constants, and optimal designs.

Loss	W	C_{GM}	Optimal information $\mathcal{I}^\star$	Optimal proportions
Interior mixed qubits (Bloch coordinates)				
Infidelity	$H/4$	$9/4$	$H/3$	$\pi_B = (1/3, 1/3, 1/3)$
Squared Bures	$H/4$	$9/4$	$H/3$	$\pi_B = (1/3, 1/3, 1/3)$
Squared Frobenius	$\mathrm{I}_3/2$	$(2+s_0)^2/2$	$H^{1/2}/(2+s_0)$	$\pi_F = \frac{(s_0,1,1)}{2+s_0}$
Squared full trace norm	I_3	$(2+s_0)^2$	$H^{1/2}/(2+s_0)$	$\pi_F = \frac{(s_0,1,1)}{2+s_0}$
Pure states				
Infidelity	$H/4$	d	$H/2$	$(1/2, 1/2)$
Squared Bures	$H/4$	d	$H/2$	$(1/2, 1/2)$
Squared Frobenius	$H/2$	$2d$	$H/2$	$(1/2, 1/2)$
Squared full trace norm	H	$4d$	$H/2$	$(1/2, 1/2)$

For mixed qubits, the fidelity weights use Miszczak et al. (2009). Section 3.1 constructs the established allocations and their information (Hou et al., 2016); the Frobenius constant is half their Bloch-vector mean-square-error constant (Eq. (4)). Squared full trace-norm loss is twice squared Frobenius loss.

For pure states, fidelity weights use Braunstein and Caves (1994, Eqs. (4), (28) and (31)). The pure-state proportions $(1/2, 1/2)$ allocate equally between the two phase branches defined in Section 4.2, equation (4.3). When aligned at the truth, the resulting measurement is Fisher-symmetric with $\mathcal{I}^\star = H/2$ by Proposition 4.1 and (4.4) (Li et al., 2016, Eqs. (16)–(17)). Squared Frobenius and squared full trace-norm losses are $2(1-F)$ and $4(1-F)$, respectively. See Theorems 4.2 and 4.3 for the information and covariance limits.

3. Adaptive tomography of mixed qubits

This section analyzes two repeatedly adaptive ridge procedures for interior mixed qubits. Section 3.1 defines the adaptive ridge estimator and the adaptive measurement axes and allocations for the Bures and Frobenius rules. Section 3.2 establishes strong consistency,

convergence to the corresponding information targets, and asymptotic normality with inverse-information covariance. Section 5 gives the loss-optimality bounds and expected physical-loss attainment.

3.1. *Ridge estimation and adaptive allocations*

Fix $\|\theta_0\| < 1$ and $\lambda > 0$. The binary law in Section 2 has conditional mean $a_t^\top \theta_0$ and variance $1 - (a_t^\top \theta_0)^2$. Define the ridge fit by (Qi et al., 2013; Mu et al., 2020)

$$\widehat{\theta}_n = \arg\min_{\theta \in \mathbb{R}^3} \left\{ \frac{1}{2} \sum_{t=1}^{n} (Y_t - a_t^\top \theta)^2 + \frac{\lambda}{2} \|\theta\|^2 \right\}.$$

At each update, the adaptive ridge estimator uses all observations available so far. The ridge keeps the problem nonsingular and its minimizer unique even before the observed axes span $\mathbb{R}^3$. With $G_n = \sum_{t \le n} a_t a_t^\top$ and $B_n = \sum_{t \le n} a_t Y_t$, the solution is $\widehat{\theta}_n = (\lambda \mathrm{I}_3 + G_n)^{-1} B_n$. The ridge fit requires one positive definite solve in three unknowns. The next design requires an estimate in the open Bloch ball; define it for every history as follows.

For the Bures rule, use

$$c_n = \widehat{\theta}_n \, \mathbf{1}(\|\widehat{\theta}_n\| < 1). \tag{3.1}$$

Here $\mathbf{1}(\cdot)$ denotes the indicator function, equal to 1 when the condition in parentheses holds and 0 otherwise. For the Frobenius rule, use the Euclidean projection

$$c_n = \Pi_{\{x : \|x\| \le R_n\}} \widehat{\theta}_n, \qquad R_n = \{1 - (n+1)^{-1/2}\}^{1/2}. \tag{3.2}$$

Here Π denotes Euclidean projection onto the indicated closed ball: $c_n = \widehat{\theta}_n$ if $\|\widehat{\theta}_n\| \le R_n$, and $c_n = R_n \widehat{\theta}_n / \|\widehat{\theta}_n\|$ otherwise.

For the oracle measurement, set $H_0 = H(\theta_0)$ and $s_0 = \{1 - \|\theta_0\|^2\}^{1/2}$. Choose a radial-first orthonormal eigenbasis $v_1, v_2, v_3 \in \mathbb{R}^3$ of the 3×3 quantum Fisher information matrix H_0, with $O_0 = (v_1, v_2, v_3)$ and $H_0 = O_0 \, \mathrm{diag}(s_0^{-2}, 1, 1) O_0^\top$. For $\theta_0 \neq 0$, take $v_1 = \theta_0 / \|\theta_0\|$; at $\theta_0 = 0$, any orthonormal triple is allowed. The Pauli matrices in Section 2.2 give the observables and their two projective measurement effects:

$$\sigma_k^{(0)} = \sum_{r=1}^{3} (v_k)_r \sigma_r, \qquad E_{k,\pm}^{(0)} = \frac{\mathrm{I}_2 \pm \sigma_k^{(0)}}{2}, \qquad k = 1, 2, 3. \tag{3.3}$$

Let $\pi = (\pi_1, \pi_2, \pi_3)$, with π_k the probability of selecting setting k. Its classical Fisher information matrix is

$$\mathcal{I}^{\pi}(\theta_0) = \sum_{k=1}^{3} \frac{\pi_k v_k v_k^\top}{1 - (v_k^\top \theta_0)^2} = O_0 \, \mathrm{diag}(\pi_1 / s_0^2, \pi_2, \pi_3) O_0^\top.$$

The Bures and Frobenius allocations are $\pi_B = (1/3, 1/3, 1/3)$ and $\pi_F(\theta_0) = (s_0, 1, 1)/(2 + s_0)$, giving $\mathcal{I}_B^\star = H_0/3$ and $\mathcal{I}_F^\star = H_0^{1/2}/(2 + s_0)$, respectively. They attain equality in (2.9) for $W_B = H_0/4$ and $W_F = \mathrm{I}_3/2$ (Gill and Massar, 2000; Hou et al., 2016). The information does not depend on the transverse pair: its two allocation probabilities are equal, and $v_2 v_2^\top + v_3 v_3^\top = \mathrm{I}_3 - v_1 v_1^\top$. At $\theta_0 = 0$, where no radial direction is distinguished, both allocations reduce to $(1/3, 1/3, 1/3)$ and yield $\mathrm{I}_3/3$ for every orthonormal triple.

For the adaptive version, at the start of batch j set $n = n_-(t) = N_{j-1}$ for $N_{j-1} < t \le N_j$. In the oracle construction replace θ_0 by c_n, H_0 by $H(c_n)$, and s_0 by $s_n = \{1 - \|c_n\|^2\}^{1/2}$. Denote the resulting axes and observables by $v_{n,k}$ and $\sigma_{n,k}$. The two-outcome PVM is $\mathsf{M}_{n,k} = \{(\mathrm{I}_2 + y\,\sigma_{n,k})/2 : y \in \{-1, 1\}\}$, where the effect indexed by y corresponds to the recorded outcome $Y_t = y$. Use allocation π_B or $\pi_F(c_n)$, holding the axes and probabilities fixed throughout the batch. The following formula specifies a choice of measurement axes for every design estimate.

Let e_1, e_2, e_3 be the standard real coordinate vectors. For $c_n \neq 0$, put $u_n = c_n/\|c_n\|$. Define the Householder completion matrix $Q(c_n)$ and the radial-first eigenbasis by (Householder, 1958)

$$Q(c_n) = \begin{cases} \mathrm{I}_3, & u_n = e_3, \\ \mathrm{I}_3 - \dfrac{2(u_n - e_3)(u_n - e_3)^{\mathsf{T}}}{\|u_n - e_3\|^2}, & u_n \neq e_3. \end{cases} \tag{3.4}$$

$$(v_{n,1}, v_{n,2}, v_{n,3}) = (u_n, Q(c_n)e_1, Q(c_n)e_2).$$

Proof of (3.4). See Supplementary Material, Section S2.5. □

The orthogonal matrix $Q(c_n)$ sends e_3 to u_n, so these are the radial and two transverse eigenvectors of $H(c_n)$. This matrix acts on real Bloch vectors and specifies measurement axes; it is not itself a qubit gate. Their projective measurements can be implemented by rotating the qubit before fixed Pauli readout, with axis signs absorbed into outcome labels. At $c_n = 0$, set $Q(0) = \mathrm{I}_3$ and use the reference triple (e_3, e_1, e_2).

For each copy t in the batch, conditionally on $\mathcal{F}_{t-1}$ draw $k_t \in [3] = \{1, 2, 3\}$ with probabilities π_1, π_2, π_3. The selected action has law

$$a_t = v_{n,k_t}, \qquad a_t \mid \mathcal{F}_{t-1} \sim \xi_t = \sum_{k=1}^{3} \pi_k \delta_{v_{n,k}}.$$

Measure M_{n,k_t} and record (Y_t, a_t). The outcomes are generated at the true state, giving the joint law

$$\mathbb{P}\{k_t = k, Y_t = y \mid \mathcal{F}_{t-1}\} = \pi_k \frac{1 + y\, v_{n,k}^{\mathsf{T}} \theta_0}{2}, \qquad k \in [3], \quad y \in \{-1, 1\}.$$

In Algorithms 1–2, $\mathcal{I}_{\xi_t}(c_n)$ is the classical Fisher information matrix evaluated at the fitted state (Section 2.4).

Algorithm 1 B: mixed-qubit Bures design

Input: the adaptive ridge estimate $\widehat{\theta}_n$ at $n = N_{j-1}$ and the next batch size m_j.

1. Compute c_n by (3.1).
2. Construct the radial-first eigenbasis of $H(c_n)$ using $Q(c_n)$ in (3.4), with the reference triple (e_3, e_1, e_2) at $c_n = 0$. Substitute $v_{n,k}$ for v_k in (3.3) to obtain the PVMs $\mathsf{M}_{n,k}$.
3. Set $\pi_{B,k} = 1/3$. For every $n < t \le n + m_j$, use $\xi_t = \sum_{k=1}^{3} \pi_{B,k} \delta_{v_{n,k}}$. Conditionally on the past, draw $k_t \sim \mathrm{Unif}([3])$ and set $a_t = v_{n,k_t}$, so $a_t \sim \xi_t$. Measure M_{n,k_t} and record (Y_t, a_t), with $Y_t \in \{-1, 1\}$. During this batch, its classical Fisher information at the fitted state is $\mathcal{I}_{\xi_t}(c_n) = H(c_n)/3$.
4. At $N_j = n + m_j$, recompute $\widehat{\theta}_{N_j}$ from all records, set $n = N_j$, and repeat for batch $j + 1$.

Algorithm 2 F: mixed-qubit Frobenius design

Input: the adaptive ridge estimate $\widehat{\theta}_n$ at $n = N_{j-1}$ and the next batch size m_j.

1. Compute c_n by (3.2) and $s_n = \{1 - \|c_n\|^2\}^{1/2}$.
2. Construct the radial-first eigenbasis of $H(c_n)$ using $Q(c_n)$ in (3.4), with the reference triple (e_3, e_1, e_2) at $c_n = 0$. Substitute $v_{n,k}$ for v_k in (3.3) to obtain the PVMs $\mathsf{M}_{n,k}$.
3. Set $(\pi_{F,1}, \pi_{F,2}, \pi_{F,3}) = (s_n, 1, 1)/(2 + s_n)$. For every $n < t \le n + m_j$, use $\xi_t = \sum_{k=1}^{3} \pi_{F,k}\delta_{v_{n,k}}$. Conditionally on the past, draw $k_t \in [3]$ with $\mathbb{P}\{k_t = k \mid \mathcal{F}_{t-1}\} = \pi_{F,k}$ and set $a_t = v_{n,k_t}$, so $a_t \sim \xi_t$. Measure M_{n,k_t} and record (Y_t, a_t), with $Y_t \in \{-1, 1\}$. During this batch, its classical Fisher information at the fitted state is $\mathcal{I}_{\xi_t}(c_n) = H(c_n)^{1/2}/(2 + s_n)$.
4. At $N_j = n + m_j$, recompute $\widehat{\theta}_{N_j}$ from all records, set $n = N_j$, and repeat for batch $j + 1$.

Algorithm 1 adapts the measurement axes with fixed allocation; Algorithm 2 adapts both. Each design is locally optimal at the fitted state; Theorem 3.1 establishes convergence to the oracle information at the truth.

3.2. *Consistency, information convergence and asymptotic normality*

Put $s_0 = \{1 - \|\theta_0\|^2\}^{1/2}$. The oracle information matrices are

$$\begin{aligned} \mathcal{I}_B^\star &= H(\theta_0)/3, \\ \mathcal{I}_F^\star &= \frac{H_0^{1/2}}{2 + s_0} = \frac{\mathrm{I}_3}{2 + s_0} + \frac{\theta_0\theta_0^\mathsf{T}}{s_0(2 + s_0)(1 + s_0)}. \end{aligned} \tag{3.5}$$

Both formulas are continuous at $\theta_0 = 0$.
The next theorem establishes consistency and convergence to the loss-optimal information. Theorem 3.2 then gives asymptotic normality.

Theorem 3.1 (Strong consistency and Fisher information convergence)**.** *Let $L = B$ denote Algorithm 1 with allocation π_B, and $L = F$ denote Algorithm 2 with allocation $\pi_F(c_{n_-(t)})$ on copy t. Both rules use the adaptive measurement axes $v_{n_-(t),k}$ defined in (3.4), with the design estimate $c_{n_-(t)}$ from the corresponding algorithm. Here $n_-(t) = N_{j-1}$ for $N_{j-1} < t \le N_j$, as defined in (2.1). For either rule $L \in \{B, F\}$, every fixed interior truth, every $\lambda > 0$, and every batch sequence (2.1), the implemented estimator satisfies almost surely*

$$\begin{aligned} &\widehat{\theta}_n \to \theta_0, \qquad c_n \to \theta_0, \qquad \mathcal{I}_{\xi_t}(\theta_0) \to \mathcal{I}_L^\star, \\ &\frac{1}{n}\sum_{t \le n} \mathcal{I}_{\xi_t}(\theta_0) \to \mathcal{I}_L^\star, \qquad \frac{1}{n}\sum_{t \le n} \mathcal{I}_{a_t}(\theta_0) \to \mathcal{I}_L^\star. \end{aligned} \tag{3.6}$$

Both safeguards, (3.1) and (3.2), are eventually inactive almost surely.

Proof of Theorem 3.1. See Supplementary Material, Section S1.2. □

With the information limit established, we next describe the estimator's Gaussian limit and its inverse-information covariance.

Theorem 3.2 (Asymptotic normality with inverse-information covariance)**.** *Under the hypotheses of Theorem 3.1, the ridge estimator and the physical output are asymptotically normal:*

$$n^{1/2}(\widehat{\theta}_n - \theta_0) \Longrightarrow \mathcal{N}_3(0, \Sigma_L), \qquad \Sigma_L = (\mathcal{I}_L^\star)^{-1}, \tag{3.7}$$

and the asymptotic normality in (3.7) also holds with c_n in place of $\widehat{\theta}_n$. Explicitly,

$$\begin{aligned} \Sigma_B &= 3(\mathrm{I}_3 - \theta_0\theta_0^{\mathsf{T}}), \\ \Sigma_F &= (2 + s_0)\left(\mathrm{I}_3 - \frac{\theta_0\theta_0^{\mathsf{T}}}{1 + s_0}\right). \end{aligned} \tag{3.8}$$

The assertions cover the maximally mixed state, single-shot updates, and all completed-batch subsequences.

Proof of Theorem 3.2. See Supplementary Material, Section S1.2. □

Comparison with earlier attainment results. Gill and Massar (2000, Theorem VII) already establish asymptotic normality and a mean-square-error expansion for a two-stage qubit procedure: an initial estimate orients the second-stage experiment toward a chosen information target. For interior mixed qubits, their second-stage likelihood estimator has an explicit frequency-inversion expression. Choosing their target scaled information to be $\mathcal{I}_L^\star$ in (3.5) yields the same limiting covariance as in (3.7). Fujiwara (2006, Theorems 8–9) proves strong consistency and inverse-information asymptotic normality for repeatedly adaptive maximum likelihood under its regularity assumptions. Fujiwara (2011) corrects the 2006 paper's separate Section 6 argument concerning achievability of the most informative Cramér–Rao bound. Qi et al. (2017) develop recursive linear regression with adaptive measurement optimization, without proving inverse-information asymptotic normality for its reported estimator.

To the best of our knowledge, earlier results do not establish these joint conclusions for the explicit adaptive ridge procedures in Algorithms 1 and 2. Both procedures refit using all available observations at every batch boundary. We prove strong consistency, convergence of the current conditional, averaged conditional, and averaged realized-setting Fisher information in (3.6) to the loss-optimal targets (3.5), and inverse-information asymptotic normality (3.7) for arbitrary predetermined positive finite batches (2.1). These conclusions include the maximally mixed state. Corollary 5.2 establishes expected-loss attainment for the physical ridge outputs.

4. Adaptive tomography of pure states

A measurement M is *Fisher symmetric* at a pure state P when its coordinate-free Fisher information form satisfies $\mathcal{J}_{\mathsf{M},P} = \mathcal{H}_P/2$ on T_P. Equivalently, $\mathcal{I}_{\mathsf{M}}(\theta) = H(\theta)/2$ whenever $P = \Gamma(\theta)$ by (2.4)–(2.6). Informational completeness provides global identification; Fisher symmetry is a local precision property.

For the pure-state losses in Table 1, W is proportional to H. Then $\mathcal{I} = H/2$ attains equality in the Gill–Massar inequality (2.8) and attains the weighted bound (2.9), giving a locally loss-optimal design among finite single-copy POVMs.

Section 4.1 introduces local coordinates, and Section 4.2 constructs a phase-paired reference measurement that is Fisher symmetric at P_{ref}. We define the ridge estimate, adaptive measurement alignment, and tangent correction. Theorem 4.2 establishes consistency and information convergence; Theorem 4.3 gives estimation rates and asymptotic normality of the tangent-corrected estimator in a fixed chart at the truth. Section 5 gives the weighted oracle bound and expected-loss attainment.

For the pure-state experiment, Z_t denotes the generic outcome O_t of Section 2; its complete label is defined below.

4.1. Charts for pure states

Fix an orthonormal basis $e_0, e_1, \ldots, e_d$ of $\mathbb{C}^D$, where $d = D - 1$, and the fixed reference projector $P_{\text{ref}} = e_0 e_0^*$. The unknown true state is P_0. For real coordinate vectors (x, y) in

$$\mathcal{B} = \{(x, y) \in \mathbb{R}^d \times \mathbb{R}^d : \|x\|^2 + \|y\|^2 < 4\},$$

define the unit-vector map and projector parametrization by

$$\begin{aligned} \psi(x, y) &= \{1 - (\|x\|^2 + \|y\|^2)/4\}^{1/2} e_0 + \frac{1}{2} \sum_{r=1}^{d} (x_r + i y_r) e_r, \\ \Gamma_{\text{ref}}(x, y) &= \psi(x, y)\psi(x, y)^*. \end{aligned} \tag{4.1}$$

Orthonormality gives $\|\psi(x, y)\| = 1$. With the unit sphere, ray space and map Φ of Definition A.1, the construction is

$$\mathcal{B} \xrightarrow{\psi} S^{2D-1} \xrightarrow{v \mapsto [v]} \mathbb{CP}^{D-1} \xrightarrow{\Phi} \mathcal{P}_D, \qquad \Gamma_{\text{ref}}(x, y) = \Phi([\psi(x, y)]).$$

Here $\Gamma_{\text{ref}}(0, 0) = P_{\text{ref}}$, and the factor $1/2$ makes $H(0) = \mathrm{I}_{2d}$.

For a fixed unitary $R = (h, b_1, \ldots, b_d)$, with $h = Re_0$ and $b_r = Re_r$, rotate this reference parametrization:

$$\Gamma_{h,b}(x, y) = R\Gamma_{\text{ref}}(x, y)R^*. \tag{4.2}$$

Thus $\Gamma_{h,b}(0, 0) = hh^*$. Appendix A, following Definition A.2, gives the domains and inverse coordinate maps.

The rotated parametrization describes measurements centered at a preliminary pure-state estimate. The asymptotic analysis uses a fixed chart at the truth; the algorithm requires neither global coordinates nor a globally smooth choice of basis for $h^\perp$.

4.2. Identification and a Fisher-symmetric reference measurement

Choose a finite reference POVM $\mathsf{F} = \{F_r : r \in \mathcal{R}\}$ with nonzero rank-one effects satisfying

$$\begin{aligned} &\text{(C1)} \quad \operatorname{span}_{\mathbb{R}}\{F_r : r \in \mathcal{R}\} = \operatorname{Herm}(\mathbb{C}^D), \\ &\text{(C2)} \quad p_r^{\text{ref}} = \operatorname{tr}(P_{\text{ref}} F_r) > 0 \quad \text{for every } r \in \mathcal{R}. \end{aligned}$$

Condition (C1) gives informational completeness on the full state space: the Born probabilities identify every density matrix, and $|\mathcal{R}| \geq D^2$ (Acharya, Kypraios and Guță, 2019; Bisio et al.,

2009). Condition (C2) gives well-defined scores near P_{ref} (Barndorff-Nielsen and Gill, 2000, Section 2); zero probabilities at other states are allowed. The label r denotes a reference outcome, while x, y remain real coordinate vectors.

We first give a reference POVM implemented by finite randomization over ordinary projective measurements. Its fixed basis changes use two-level rotations and single-level phase gates, which multiply one basis vector by i, $-i$, or -1 and leave the others unchanged.

Example 4.1 (Randomized-PVM implementation of the reference POVM)**.** Choose B fixed orthonormal bases $\{f_{\ell,k} : k \in [D]\}$, $\ell \in [B]$, whose projectors span $\mathrm{Herm}(\mathbb{C}^D)$ and satisfy $e_0^* f_{\ell,k} \neq 0$. Set

$$\mathcal{R} = [B] \times [D], \qquad F_{\ell,k} = \frac{1}{B} f_{\ell,k} f_{\ell,k}^*.$$

Select ℓ_t uniformly, perform its PVM, and record $r_t = (\ell_t, k_t) \in \mathcal{R}$, where $k_t \in [D]$ is the outcome of the selected PVM; the effects include the basis probability $1/B$. Necessarily $B \geq D + 1$, since the projectors span at most $1 + B(D-1)$ real Hermitian directions. This bound alone does not ensure (C1), even when $B = D + 1$; in particular, $B = 1$ is impossible for $D \geq 2$.

Proof of Example 4.1. See Supplementary Material, Section S2.5. The explicit reference bases are given in Proposition S5.1. □

Conditions (C1)–(C2) give identification and positive reference probabilities. We next balance real and imaginary tangent information to attain the loss-optimal target $\mathcal{H}_{P_{\mathrm{ref}}}/2$.

Phase pairing. For any rank-one reference POVM F satisfying (C1)–(C2), define

$$\begin{aligned} V &= \mathrm{I}_D + (i-1)P_{\mathrm{ref}}, \\ E_{0,r} &= F_r/2, \qquad E_{1,r} = V^* F_r V/2. \end{aligned} \tag{4.3}$$

Write $z = (b, r) \in \mathcal{Z} := \{0,1\} \times \mathcal{R}$, $E_z = E_{b,r}$, and $\mathsf{E} = \{E_z : z \in \mathcal{Z}\}$. The gate V applies a $+\pi/2$ phase to e_0 and leaves $e_0^\perp$ unchanged. Draw a fair branch b_t, apply V^{b_t} before measuring F, and retain $Z_t = (b_t, r_t) \in \mathcal{Z}$. At a pure state P,

$$\mathbb{P}\{Z_t = z\} = \tfrac{1}{2} \operatorname{tr}\big(P(V^b)^* F_r V^b\big) = \operatorname{tr}(PE_z).$$

The complete effects include the branch probability $1/2$ and, in Example 4.1, the basis probability $1/B$. Each branch remains informationally complete. The PVM implementation uses classical randomization, projective readout, and an optional $\pi/2$ phase shift, given the required basis and phase controls. Li et al. (2016, Eq. (18)) use a fair choice between two phase-related orthonormal bases.

Here $\mathcal{I}_{\mathsf{E}}(0)$ is the coordinate Fisher matrix of the paired POVM at $P_{\mathrm{ref}} = \Gamma_{\mathrm{ref}}(0,0)$ in (4.1), as defined in (2.4).

The next proposition shows that phase pairing preserves informational completeness and yields Fisher symmetry at the reference state.

Proposition 4.1 (Phase pairing)**.** *The effects* (4.3) *form an informationally complete measurement on the full state space and have positive reference probabilities. In the reference coordinates* (4.1)*, centered at* P_{ref}*,*

$$\mathcal{I}_{\mathsf{E}}(0) = H(0)/2 = \mathrm{I}_{2d}/2 \tag{4.4}$$

Equivalently, $\mathcal{J}_{\mathsf{E},P_{\mathrm{ref}}} = \frac{1}{2}\mathcal{H}_{P_{\mathrm{ref}}}$ *as bilinear forms on* $T_{P_{\mathrm{ref}}}$*; this is Fisher symmetry at* P_{ref}*.*

Proof of Proposition 4.1. See Supplementary Material, Section S2.4. □

Remark 1. Let $\mathcal{I}_{\mathsf{F}}(0)$ be the Fisher matrix (2.4) in coordinates (4.1). Conditions (C1)–(C2) alone do not ensure $\mathcal{I}_{\mathsf{F}}(0) = H(0)/2$. Pairing (4.3) balances real and imaginary tangent information to enforce (4.4); it is unnecessary if F is already Fisher symmetric.

Proof of Remark 1. See Supplementary Material, Section S2.4. □

A fixed finite-outcome, single-copy measurement cannot be Fisher symmetric at every pure state (Zhu and Hayashi, 2018, Theorem 1). We therefore use a preliminary estimate to rotate this reference measurement toward the unknown state.

4.3. Preliminary least-squares estimation

On copy t, let U_t be a unitary chosen from past observations, with its adaptive choice specified in Section 4.4. The measurement effects and observed categorical outcome are

$$E_{t,z} = U_t E_z U_t^*, \qquad z \in \mathcal{Z}, \qquad Z_t = (b_t, r_t) \in \mathcal{Z}.$$

Here b_t records the fair coin selecting the phase branch, and r_t is the outcome of the adaptively aligned measurement. For $\lambda > 0$, define the preliminary estimator from all n records by

$$\widehat{A}_n = \operatorname*{arg\,min}_{A \in \mathrm{Herm}(\mathbb{C}^D)} \left\{ \frac{\lambda}{2} \|A\|_F^2 + \frac{1}{2} \sum_{t \le n} \sum_{z \in \mathcal{Z}} \left[\mathbf{1}_{\{Z_t = z\}} - \mathrm{tr}(A E_{t,z}) \right]^2 \right\}. \tag{4.5}$$

This is ordinary ridge least squares with a Hermitian matrix parameter (Qi et al., 2013; Mu et al., 2020). No positivity, trace-one, or rank-one constraint is imposed on $\widehat{A}_n$. Since $\lambda > 0$, the criterion is strictly convex and has a unique minimizer for every record.

Project the raw estimate onto the pure states by taking

$$\widehat{h}_n \in \operatorname*{arg\,max}_{\|h\|=1} h^* \widehat{A}_n h, \qquad \widehat{P}_n = \widehat{h}_n \widehat{h}_n^*.$$

Use fixed tie and phase conventions for the largest-algebraic-eigenvalue unit eigenvector $\widehat{h}_n$. Then $\widehat{P}_n$ is a nearest rank-one projector to $\widehat{A}_n$ in Frobenius norm, even when the raw estimate is not positive semidefinite; the minimizing property is used in the proof of Theorem 4.2.

Condition (C1) also yields a uniform positive lower bound on the unregularized least-squares curvature under every unitary alignment; the frame-operator calculation is used in the proof of Theorem 4.2.

4.4. Adaptive measurement alignment

The paired reference POVM E is Fisher symmetric at P_{ref}. To align it with a candidate state aa^*, use the unit-vector action space $\mathcal{A} = S^{2D-1}$. For $a \in \mathcal{A}$, the phase-adjusted Householder unitary is (Householder, 1958; LAPACK, 2026)

$$\begin{aligned} \gamma(a) &= \begin{cases} e_0^* a/|e_0^* a|, & e_0^* a \neq 0, \\ 1, & e_0^* a = 0, \end{cases} \\ v(a) &= e_0 + \overline{\gamma(a)} a, \\ U(a) &= \gamma(a) \left(\frac{2v(a)v(a)^*}{v(a)^* v(a)} - \mathrm{I}_D \right). \end{aligned} \tag{4.6}$$

Here $v(a)^* v(a) = 2(1 + |e_0^* a|) \in [2, 4]$ and $U(a)^* U(a) = \mathrm{I}_D$. The prefactor $\gamma(a)$ ensures $U(a)e_0 = a$ exactly, as well as $U(a)P_{\mathrm{ref}}U(a)^* = aa^*$.

Proof of (4.6). See Supplementary Material, Section S2.5. □

Taking $R = U(a)$ in (4.2) centers the parametrization at aa^*. Thus $U(a)$ rotates e_0 to a, while the phase-adjusted Householder gate $U(a)^*$ sends the candidate vector a and state aa^* back to the reference vector e_0 and state P_{ref}, respectively. The measurement associated with action a is

$$\mathsf{E}_a = \{E_{a,z} : z \in \mathcal{Z}\}, \qquad E_{a,z} = U(a) E_z U(a)^*.$$

Thus a is the statistical action, and $U(a)$ implements *measurement alignment* by rotating the reference POVM E to E_a and mapping P_{ref} to the candidate state aa^*. Unitary conjugation transforms the classical and quantum Fisher information forms in the same way, so Proposition 4.1 implies Fisher symmetry of E_a at aa^*.

At the start of batch j, put $n = N_{j-1}$. Here $\widehat{h}_n$ is a unit eigenvector for the largest algebraic eigenvalue of the ridge estimate $\widehat{A}_n$ fitted to the first n records in (4.5), and $\widehat{P}_n = \widehat{h}_n \widehat{h}_n^*$, as in step 1 of Algorithm 3. For the predetermined batches (2.1), set

$$a_t = \widehat{h}_n, \qquad \xi_t = \delta_{a_t}, \qquad U_t = U(a_t), \qquad N_{j-1} < t \leq N_j.$$

Here a_t is the measurement alignment vector estimated from the preceding data; physically, applying U_t^* sends this estimated direction to e_0 before the reference measurement. The point mass $\xi_t = \delta_{a_t}$ holds the alignment fixed within the batch, and the next batch uses the estimate updated with the new outcomes. The effects on copy t are $E_{t,z} = E_{a_t,z}$, and

$$\mathbb{P}\{Z_t = z \mid \mathcal{F}_{t-1}\} = \mathrm{tr}(P_0 E_{t,z}), \qquad z \in \mathcal{Z}.$$

Since $\xi_t = \delta_{a_t}$, the notation in (2.4) gives $\mathcal{J}_{\xi_t,P} = \mathcal{J}_{a_t,P}$ and $\mathcal{I}_{\xi_t}(\theta) = \mathcal{I}_{a_t}(\theta)$. The first equality concerns coordinate-free forms; the second concerns their matrices in the same fixed parametrization.

To implement action a_t, apply U_t^* to a fresh copy and perform the phase-paired reference measurement of Section 4.2. Store (Z_t, a_t), with $Z_t = (b_t, r_t)$ and $U_t = U(a_t)$. The rule applies to any rank-one reference POVM satisfying (C1)–(C2). Example 4.1 gives a randomized-PVM implementation.

Theorem 4.2 proves that $a_t a_t^* \to P_0$ and $\mathcal{J}_{a_t,P_0} \to \mathcal{H}_{P_0}/2$, without requiring convergence of every column of U_t. Every aligned measurement remains informationally complete, as used in its proof. To attain the inverse-information covariance, we next correct the preliminary fit.

4.5. A weighted tangent correction

Optimal information does not ensure efficient least squares. Root-n consistency of the preliminary projection does not imply efficiency. For the finite measurement specified in the proof below, with ridge coefficient $\lambda = 1$, the adaptive information converges to $\mathcal{H}_{P_0}/2$, but its preliminary estimator satisfies

$$n^{1/2}(\widehat{P}_n - P_0) \Longrightarrow \mathcal{N}_T(0, v_D \mathrm{I}_T),$$
$$v_D = \frac{4(D-1)(5D-7)}{(4D-5)^2} = 1 + \frac{(2D-3)(2D-1)}{(4D-5)^2} > 1. \tag{4.7}$$

Here $T = T_{P_0}$ is the true tangent space with its Frobenius inner product; the inverse-information covariance is I_T, since the coordinate-free quantum Fisher information form $\mathcal{H}_{P_0}$ is twice the Frobenius inner product on T. For $D = 2$, the uncorrected variance is $4/3$ in each tangent direction, one third above the available optimum. Here unequal outcome probabilities make the unweighted residual fit inefficient.

Proof of (4.7). See Supplementary Material, Section S2.6, using the measurement construction in Section S8. □

To recover the available information, we correct the preliminary fit on its tangent space. For an outcome encoded as a one-hot vector, the noise covariance is $\mathrm{diag}(p) - pp^{\mathsf{T}}$.

Fix a reporting sample size n and abbreviate $h = \widehat{h}_n$ and $P = \widehat{P}_n = hh^*$, using the preliminary estimates in Section 4.3. We suppress n in the fitted quantities below; sums over t, z run over $t \le n$ and $z \in \mathcal{Z}$. Let $\Pi_P(A) = PA(\mathrm{I} - P) + (\mathrm{I} - P)AP$ be the orthogonal projection onto the tangent space T_P defined in Section 2; Definition A.5 describes this projection. The real dimension is $2d$, since the local parametrization (4.2) uses $(x, y) \in \mathbb{R}^{2d}$, with d real coordinates in each of x and y. With $E_{t,z} = E_{a_t,z}$ denoting the complete labeled measurement, set

$$q_{t,z} = \mathrm{tr}(PE_{t,z}) = h^* E_{t,z} h, \qquad r_{t,z} = \mathbf{1}_{\{Z_t = z\}} - q_{t,z},$$
$$\tau = \tfrac{1}{4} \min_z \mathrm{tr}(P_{\mathrm{ref}} E_z) > 0, \qquad \bar{q}_{t,z} = \max(q_{t,z}, \tau).$$

Each $q_{t,z}$ evaluates the recorded measurement at the same fitted state P. Its true-state counterpart is $p_{t,z} = \mathrm{tr}(P_0 E_{t,z})$, used in the proof of Theorem 4.3. The known reference probabilities determine τ. Truncation prevents unstable denominators; the values $\bar{q}_{t,z}$ need not sum to one and are not a replacement sampling distribution.

The inverse-covariance weighting underlying this correction is the generalized least-squares principle used in quantum tomography (Acharya, Kypraios and Guță, 2019, Section 5). Our procedure applies this principle on the fitted pure-state tangent space under adaptive measurements; the probability clipping above is a finite-sample stabilization used in our construction.

Use the normalization retraction $R_P(\Delta)$ of Definition A.6 to describe nearby pure states, with first-order perturbation $P + \Delta$, where $\Delta \in T_P$. A local constrained likelihood fit solves

$$\operatorname*{arg\,max}_{\Delta \in T_P,\ \Delta \text{ near } 0} \sum_{t,z} \mathbf{1}_{\{Z_t = z\}} \log \mathrm{tr}\{E_{t,z} R_P(\Delta)\},$$

with the recorded setting matrices $E_{t,z}$ held fixed. The outcome probabilities satisfy

$$\mathrm{tr}\{E_{t,z} R_P(\Delta)\} = q_{t,z} + \mathrm{tr}(E_{t,z}\Delta) + O(\|\Delta\|_F^2).$$

For positive fitted probabilities, Fisher scoring replaces the observed likelihood curvature by its expected value at P, giving the quadratic problem

$$\arg\min_{\Delta\in T_P} \frac{1}{2}\sum_{t,z}\frac{\left[\mathbf{1}_{\{Z_t=z\}} - q_{t,z} - \mathrm{tr}(E_{t,z}\Delta)\right]^2}{q_{t,z}}.$$

This gives the inverse-probability weighted tangent fit; our estimator adds a ridge penalty and uses the clipped denominators $\bar{q}_{t,z}$ defined above for finite-sample stability. This one-step construction follows the general methodology of Brouste, Dutang and Noutsa Mieniedou (2021).

For fixed $\eta > 0$, solve

$$\widehat{\Delta}_n = \arg\min_{\Delta\in T_P}\left\{\frac{\eta}{2}\|\Delta\|_F^2 + \frac{1}{2}\sum_{t,z}\frac{(r_{t,z} - \mathrm{tr}(E_{t,z}\Delta))^2}{\bar{q}_{t,z}}\right\}. \tag{4.8}$$

The positive ridge makes this quadratic minimizer unique. Report

$$\widehat{h}_n^{\mathrm{eff}} = \frac{h + \widehat{\Delta}_n h}{\|h + \widehat{\Delta}_n h\|}, \qquad \widehat{P}_n^{\mathrm{eff}} = \widehat{h}_n^{\mathrm{eff}}(\widehat{h}_n^{\mathrm{eff}})^*. \tag{4.9}$$

Since $h^*\widehat{\Delta}_n h = 0$, the squared denominator is $1 + \|\widehat{\Delta}_n h\|^2 \geq 1$. The resulting projector is well defined and does not depend on the phase chosen for h. The normalization is a retraction (Definition A.6): its first-order change is $\widehat{\Delta}_n$, with a quadratic remainder.

The normal equations for (4.8) give a single $2d$-dimensional linear system in a Frobenius-orthonormal tangent basis.

Role of the correction. The correction removes the first-order preliminary error, leaving inverse information times the score, with a quadratic geometric remainder. The preliminary fit, tangent space and probability weights use the same adaptive records. The proof of Theorem 4.3 handles this dependence through martingale bounds and an exact reciprocal expansion, proving the cancellation with an $O_p(n^{-1})$ remainder. This is negligible at the $n^{-1/2}$ scale and yields the Gaussian limit in Theorem 4.3.

Algorithm 3 collects the measurement rule and both estimation steps. Its reported correction uses all setting–outcome records $(Z_1, a_1), \ldots, (Z_n, a_n)$, where each pair contains the complete outcome Z_t and the measurement alignment vector a_t used on copy t.

Algorithm 3 Phase-paired adaptive tomography and tangent correction

Input. A finite rank-one reference POVM $\mathsf{F} = \{F_r : r \in \mathcal{R}\}$ satisfying (C1)–(C2); the paired POVM E in (4.3); fixed $\lambda, \eta > 0$; and positive finite batches (2.1). Fix deterministic eigenvector tie and phase rules, including the initial choice $\widehat{h}_0$ when $\widehat{A}_0 = 0$.

1. **Fit the preliminary state.** At $n = N_{j-1}$, compute the ridge estimate $\widehat{A}_n$ in (4.5) from all previous records. Select its largest-algebraic-eigenvalue unit eigenvector $\widehat{h}_n$ and put $\widehat{P}_n = \widehat{h}_n \widehat{h}_n^*$.
2. **Align the next batch.** For $N_{j-1} < t \le N_j$, set $a_t = \widehat{h}_n$, $\xi_t = \delta_{a_t}$ and $U_t = U(a_t)$ using (4.6).
3. **Measure and record.** On each fresh copy, perform E_{a_t} as described in Section 4.4. Store (Z_t, a_t), with $Z_t = (b_t, r_t) \in \mathcal{Z}$ and $r_t \in \mathcal{R}$. For Example 4.1, $r_t = (\ell_t, k_t)$ records the selected reference basis $\ell_t \in [B]$ and PVM outcome $k_t \in [D]$.
4. **Refit and repeat.** At the next batch boundary, retain all accumulated records. Return to step 1 for further data, or proceed to step 5 at a reporting sample size. Future measurements continue to use the preliminary estimate.
5. **Report the corrected estimate.** At $n = N_J$, refit $\widehat{A}_n$ and $\widehat{P}_n$ from all n records. Solve (4.8). Apply the normalization (4.9) and report $\widehat{P}_n^{\mathrm{eff}}$.

Design update and reported estimate. The preliminary estimate alone selects future measurements. The weighted tangent solve and normalization are computed only at a reporting sample size. Theorems 4.2–4.3 and Corollary 5.3 give the resulting limits.

4.6. Consistency, information convergence and asymptotic normality

The next theorem establishes consistency and information convergence for the pure-state procedure. Theorem 4.3 then gives its estimation rates and limiting Gaussian law.

Theorem 4.2 (Strong consistency and Fisher information convergence)**.** *Fix $D \geq 2$, a reference POVM satisfying (C1)–(C2) with rank-one effects, $\lambda, \eta > 0$, a pure truth P_0, and batches (2.1). The paired adaptive experiment and estimators in Algorithm 3, defined by (4.5) and (4.8), satisfy, almost surely,*

$$\begin{aligned} \widehat{A}_n \to P_0, \quad \widehat{P}_n \to P_0, \quad \widehat{P}_n^{\mathrm{eff}} \to P_0, \quad & (n = N_j,\ j \to \infty), \\ U_t P_{\mathrm{ref}} U_t^* \to P_0, \qquad \mathcal{J}_{a_t, P_0} \to \mathcal{H}_{P_0}/2, \qquad & (t \to \infty), \\ \frac{1}{n} \sum_{t \leq n} \mathcal{J}_{a_t, P_0} \to \mathcal{H}_{P_0}/2, \qquad & (n = N_j,\ j \to \infty). \end{aligned} \tag{4.10}$$

Here $t \to \infty$ runs through all measured copies. The limits indexed by n hold along the completed-batch sample sizes $n = N_j$, $j \to \infty$, where N_j is the cumulative number of copies after batch j defined in (2.1).

Proof of Theorem 4.2. See Supplementary Material, Section S1.3. □

We next give estimation rates for the matrix fit and its rank-one projection, followed by asymptotic normality of the tangent-corrected output.

Theorem 4.3 (Estimation rates and asymptotic normality)**.** *Under the hypotheses of Theorem 4.2, the estimators in Algorithm 3 satisfy the following conclusions along $n = N_j$, $j \to \infty$.*
(i) Estimation rates. *The matrix least-squares estimator and its rank-one projection satisfy $\|\widehat{A}_n - P_0\|_F = O_p(n^{-1/2})$ and $\|\widehat{P}_n - P_0\|_F = O_p(n^{-1/2})$.*
(ii) Asymptotic normality. *In any fixed smooth pure-state chart $(\mathcal{U}, \varphi)$ as in Definition A.2, with $P_0 \in \mathcal{U}$, let $\theta_0 = \varphi(P_0)$ and let $H(\theta_0)$ be the matrix in the parametrization $\Gamma(\theta) = \varphi^{-1}(\theta)$. Then*

$$n^{1/2}\{\varphi(\widehat{P}_n^{\mathrm{eff}}) - \varphi(P_0)\} \Longrightarrow \mathcal{N}_{2d}(0, 2H(\theta_0)^{-1}). \tag{4.11}$$

Equivalently, $n^{1/2}(\widehat{P}_n^{\mathrm{eff}} - P_0)$ converges in distribution to a centered Gaussian supported on the true tangent space, with identity covariance in Frobenius-orthonormal tangent coordinates.

Proof of Theorem 4.3. See Supplementary Material, Section S1.3, under "Proof of Theorem 4.3". The loss distributions are in Corollary S1.1. □

Remark 2 (Comparison with earlier adaptive estimation)**.** The fitted tangent correction has a linear-system solution, avoiding iterative likelihood maximization on the pure-state manifold; local iterative methods need not reach a global maximizer (Boumal, 2023, Chapter 4). The adaptive likelihood theory (Fujiwara, 2006, 2011) and recursive regression procedures (Qi et al., 2017) discussed in Section 3.2 do not establish the stated Gaussian limit for this fitted tangent correction. For pure states, Girotti, Godley and Guță (2024) attain the Holevo bound using preliminary localization and a final displaced-null stage, while Vargas, Delgado and Pereira (2026) use a three-stage Fisher-symmetric protocol with finite-sample bounds and likelihood refinement, as discussed in Section 1. Theorems 4.2 and 4.3 jointly establish

consistency, Fisher-symmetric information convergence and the Gaussian limit (4.11) for Algorithm 3 under repeated updates in every fixed dimension and arbitrary predetermined positive finite batches, including single-copy updates.

Domain of the chart. Let $(\mathcal{U}, \varphi)$ be the fixed chart in Theorem 4.3, as defined in Definition A.2. Both estimated projectors are valid pure states for every record, but may initially fall outside $\mathcal{U}$. The almost-sure consistency in (4.10) implies that both eventually lie in $\mathcal{U}$ almost surely, and hence

$$\mathbb{P}\{\widehat{P}_n \notin \mathcal{U} \text{ or } \widehat{P}_n^{\mathrm{eff}} \notin \mathcal{U}\} \longrightarrow 0.$$

For (4.11), define $\varphi(\widehat{P}_n^{\mathrm{eff}}) = \varphi(P_0)$ on the event $\widehat{P}_n^{\mathrm{eff}} \notin \mathcal{U}$. This extension makes the coordinate random vector defined everywhere and does not change its limiting distribution. As in the theorem, n ranges over completed batches.

5. Adaptive Gill–Massar bound and expected-loss attainment

Let

$$\mathfrak{M}_D = \bigcup_{m\geq 1} \left\{ (M_1, \ldots, M_m) : M_j \succeq 0, \quad \sum_{j=1}^{m} M_j = \mathrm{I}_D \right\}$$

be the class of all finite single-copy POVMs on $\mathbb{C}^D$, with $D \times D$ effects M_j. As in Section 2, let O_i denote the measurement outcome on copy i and o_i its realized value. A prescribed, parameter-independent policy based on past records is a function ζ_t assigning to each history $\mathsf{h} = (o_1, \ldots, o_{t-1})$ a POVM $\zeta_t(\mathsf{h}) = (M_{t,o}(\mathsf{h}))_o \in \mathfrak{M}_D$, where $M_{t,o}(\mathsf{h})$ is the effect associated with outcome o. The selected POVM $\mathsf{M}_t = \zeta_t(O_1, \ldots, O_{t-1})$ is $\mathcal{F}_{t-1}$-measurable, and

$$p_{t,\theta}(o \mid \mathsf{h}) = \mathrm{tr}\{\rho(\theta) M_{t,o}(\mathsf{h})\}.$$

Here $\rho(\theta)$ is the Bloch parametrization (2.2) ($D = 2$), with true interior mixed-qubit state $\rho_0 = \rho(\theta_0)$. For pure states ($D \geq 2$), $\rho(\theta) = \Gamma(\theta)$ is a smooth local parametrization near the true state $P_0 = \Gamma(\theta_0)$, as in Definition A.2.

The *conditional POVM* $\zeta_t(\mathsf{h})$ specifies the next measurement given h. Finite randomization is included by weighting effects with selection probabilities and recording both the setting and readout in O_t. This class includes the measurements used by our algorithms.

A finite-valued coordinate estimator T_n is *locally unbiased at* θ_0 when its mean $m_n(\theta) = \mathbb{E}_\theta T_n$ satisfies $m_n(\theta_0) = \theta_0$ and has the identity Jacobian at θ_0.

Theorem 5.1 (Adaptive Gill–Massar bound)**.** *For either model above, let $H = H(\theta_0) \succ 0$ be the SLD quantum Fisher information matrix and $W \succeq 0$ a weight in the same coordinates. For any sequence of policies $(\zeta_t)_{t\geq 1}$ as above, under the conditional-likelihood assumptions in Appendix B, every finite-valued coordinate estimator T_n locally unbiased at θ_0 and every $n \geq 1$ satisfy*

$$n\, \mathbb{E}_{\theta_0}\big[(T_n - \theta_0)^{\mathsf{T}} W (T_n - \theta_0)\big] \geq C_{\mathrm{GM}}(W, H), \tag{5.1}$$

where C_{GM} is defined in (2.9).

Proof of Theorem 5.1. See Supplementary Material, Section S1.1. □

For each fixed past record h, the Fisher information of the single-copy POVM $\zeta_t(\mathsf{h})$ satisfies the Gill–Massar inequality (2.8) (Gill and Massar, 2000, Theorems I and II).

Recall from Table 1 that the squared Bures and Frobenius loss weights are $(W_B, W_F) = (H/4, H/2)$ for pure states and $(H/4, \mathrm{I}_3/2)$ for mixed qubits. The corresponding oracle constants are $(d, 2d)$ and $(9/4, (2+s_0)^2/2)$, respectively.

Corollary 5.2 (Mixed-qubit expected-loss attainment)**.** *Under the hypotheses of Theorem 3.2, let c_n be the safeguarded output (3.1) for Algorithm 1 or the projected output (3.2) for Algorithm 2. Along $n = N_j$, $j \to \infty$,*

$$\begin{aligned} n\,\mathbb{E}_{\theta_0} D_B^2\{\rho(c_n), \rho(\theta_0)\} &\longrightarrow 9/4 \qquad \textit{(Algorithm 1)}, \\ n\,\mathbb{E}_{\theta_0} \|\rho(c_n) - \rho(\theta_0)\|_F^2 &\longrightarrow (2+s_0)^2/2 \quad \textit{(Algorithm 2)}. \end{aligned} \tag{5.2}$$

Proof of Corollary 5.2. See Supplementary Material, Section S1.4. □

Corollary 5.3 (Pure-state expected-loss attainment)**.** *Under the hypotheses of Theorem 4.3, the reported tangent-corrected projector of Algorithm 3 satisfies, along $n = N_j$, $j \to \infty$,*

$$\begin{aligned} n\,\mathbb{E}_{P_0} D_B(\widehat{P}_n^{\mathrm{eff}}, P_0)^2 &\longrightarrow d, \\ n\,\mathbb{E}_{P_0} \left\| \widehat{P}_n^{\mathrm{eff}} - P_0 \right\|_F^2 &\longrightarrow 2d. \end{aligned} \tag{5.3}$$

Proof of Corollary 5.3. See Supplementary Material, Section S1.4. □

Corollaries 5.2 and 5.3 attain the squared Bures and Frobenius constants in Theorem 5.1, equation (5.1), at each fixed truth and dimension. Their Gaussian limits and the proved fourth-moment bounds give uniform integrability and hence convergence of expected loss, without additional moment assumptions. The physical outputs may be biased; local unbiasedness defines the finite-sample lower-bound comparison class, not a requirement on these outputs. This is pointwise attainment of the single-copy oracle benchmark. For regular biased comparators, Gill and Massar (2000, Theorem IV) require locally uniform convergence of scaled mean-square-error matrices, which is not claimed here. The loss identities in Section 2.5 and the corollary proofs also give the Table 1 infidelity and squared full trace-norm constants: Algorithms 1 and 2, respectively, for mixed qubits, and Algorithm 3 for pure states.

6. Simulation study

We examine loss-specific measurement design for mixed qubits and the effect of the tangent correction for pure states. All observations follow the ideal Born model, with ridge coefficients equal to one. Starting with $N_0 = 0$, we use the nominal batch schedule

$$m_j = \max\left\{16, \left\lceil \frac{N_{j-1}}{50} \right\rceil \right\}, \qquad N_j = N_{j-1} + m_j, \qquad j = 1, 2, \ldots.$$

Here N_j is the cumulative number of copies after batch j. The proportional block rule $\lceil N/50 \rceil$ is used by Struchalin et al. (2016, Section II.C); we impose a 16-copy minimum. Only the final batch is shortened to reach $16{,}384 = 2^{14}$ copies, giving 202 batches in total. We report at completed batches nearest 256, 512, and 1024, 2048, . . . , 16,384 copies; ties select the earlier endpoint. Both the horizontal and vertical axes are linear. Each allocation or measurement procedure uses $R = 10{,}000$ independent Monte Carlo replicates. The configurations and reproduction instructions accompany the archived simulation code.

For a fixed truth ρ_0 and its local loss weight W_0 from Table 1, the design panels show

$$C_n^{\mathrm{cur}} = \mathrm{tr}\Big(W_0 \mathcal{I}_{\xi_{n+1}}(\rho_0)^{-1}\Big), \qquad C_n^{\mathrm{avg}} = \mathrm{tr}\left[W_0 \left\{\frac{1}{n}\sum_{t\le n} \mathcal{I}_{\xi_t}(\rho_0)\right\}^{-1}\right]. \tag{6.1}$$

Here C_n^{cur} evaluates the next batch's conditional experiment, whereas C_n^{avg} uses the batch-size-weighted information actually collected. Figure 1 evaluates these criteria on separate adaptive records using exact finite Born sums. Figure 2(a) reports their Monte Carlo means $\widehat{C}_n^{\ell} = R^{-1}\sum_{r=1}^{R} C_n^{\ell,(r)}$, $\ell \in \{\mathrm{cur}, \mathrm{avg}\}$, over the same replicates as its loss panel, with shaded bands of 1.96 Monte Carlo standard errors. Truth enters sampling and diagnostic evaluation; selection and estimation use only observed records. The loss panels report Monte Carlo estimates of the scaled risks,

$$\widehat{R}_L(n) = \frac{n}{R}\sum_{r=1}^{R} L(\widehat{\rho}_n^{(r)}, \rho_0),$$

where $L(A, B) = \|A - B\|_F^2$ for squared Frobenius loss or $L(A, B) = D_B(A, B)^2$ for squared Bures loss, and $\widehat{\rho}_n^{(r)}$ is the output from Monte Carlo replicate r. Error bars are 1.96 Monte Carlo standard errors. Monte Carlo replicates are reused across reporting sizes, so their Monte Carlo errors are correlated. The Gill–Massar lines give information lower bounds and the corresponding asymptotic expected-loss benchmarks listed in Table 1.

6.1. Mixed-qubit design and loss

At $\theta_0 = 0.9(1,2,3)^\mathsf{T}/\sqrt{14}$, we run Algorithms 1 and 2, using their exact axes (3.4) and safeguards. Figure 1 compares the loss-matched allocation with the opposite rule. The benchmarks from Table 1 are approximately 2.9668 for squared Frobenius loss and 2.25 for squared Bures loss.

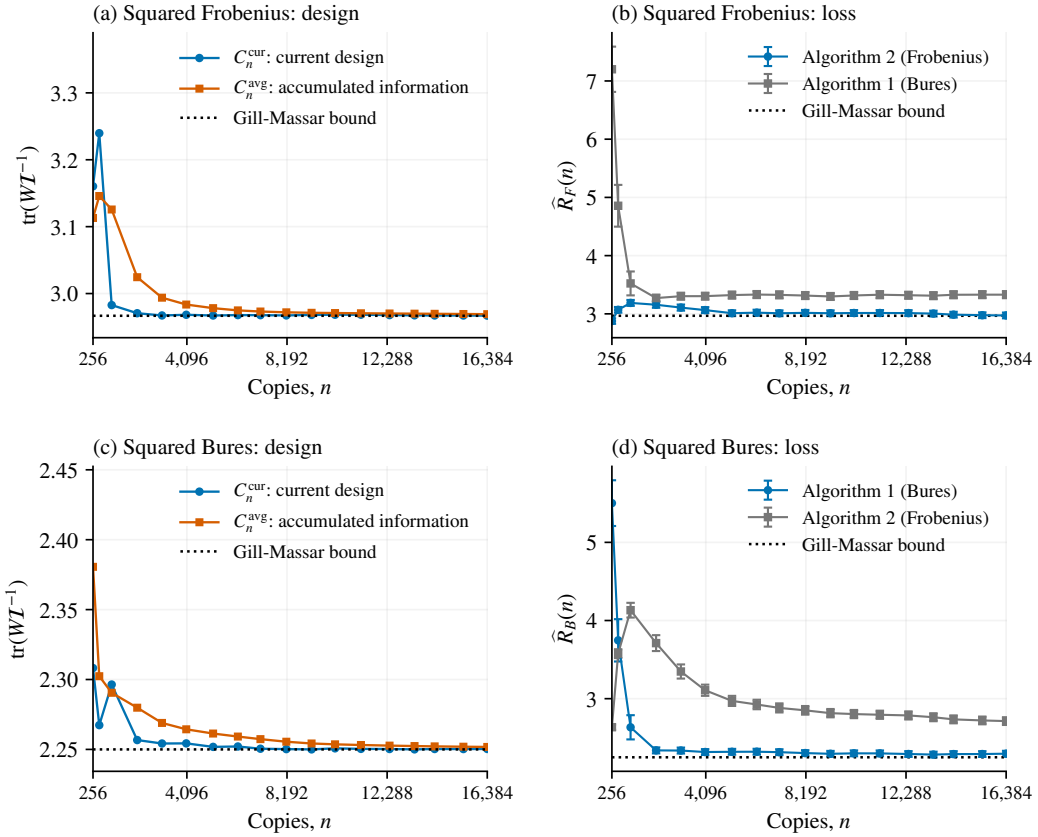


FIG 1. *Mixed-qubit design and physical loss at* $\|\theta_0\| = 0.9$. *(a,b) Squared Frobenius loss; (c,d) squared Bures loss. (a,c)* C_n^{cur} *and* C_n^{avg} *from* (6.1) *along one realized adaptive measurement record per panel. (b,d)* $\widehat{R}_L(n)$ *from 10,000 Monte Carlo replicates per allocation, with* 1.96 *Monte Carlo standard-error bars. Blue is loss-matched: Algorithm 2 in (b) and Algorithm 1 in (d); gray uses the opposite rule. Dotted lines are the Gill–Massar benchmarks.*

Panels (a) and (c) each show one realized adaptive measurement record. Theorem 3.1 gives almost-sure convergence of both criteria to the respective loss-specific Gill–Massar bound; the curves illustrate this pathwise behavior, with finite-sample fluctuations.

At $n = 16{,}384$, the matched means are 2.9726 (MC s.e. 0.0252) and 2.2961 (MC s.e. 0.0188), respectively. The Bures mean remains about 2.45 Monte Carlo standard errors above $9/4$, indicating a residual finite-sample gap. The different loss curves show why consistency alone does not determine the choice of allocation.

6.2. *Pure-state design and tangent correction*

We implement Algorithm 3 at $D = 4$ (two qubits), with $h_{0,j} = \sqrt{(j+1)/10}\, e^{0.37ij}$, $j = 0, \ldots, 3$, where i is the imaginary number. The phase-paired implementation of Example 4.1 has $104 = 2{\times}13{\times}4$ complete outcomes: two phase branches (Li et al., 2016, Eq. (18)), 13 bases built from generalized Gell–Mann operators (Bertlmann and Krammer, 2008, Section 3.1), and four readouts. Each copy uses one randomly selected four-outcome projective measurement. We initialize $\widehat{h}_0 = e_0$ and use fixed eigenvector conventions; the probability floor is $1/6240$. Only the all-record preliminary fit selects subsequent alignments. Both its rank-one projection and the reported weighted tangent correction use the same acquired records. The pure-state loss weights are proportional, so Figure 2 displays only squared Bures loss.

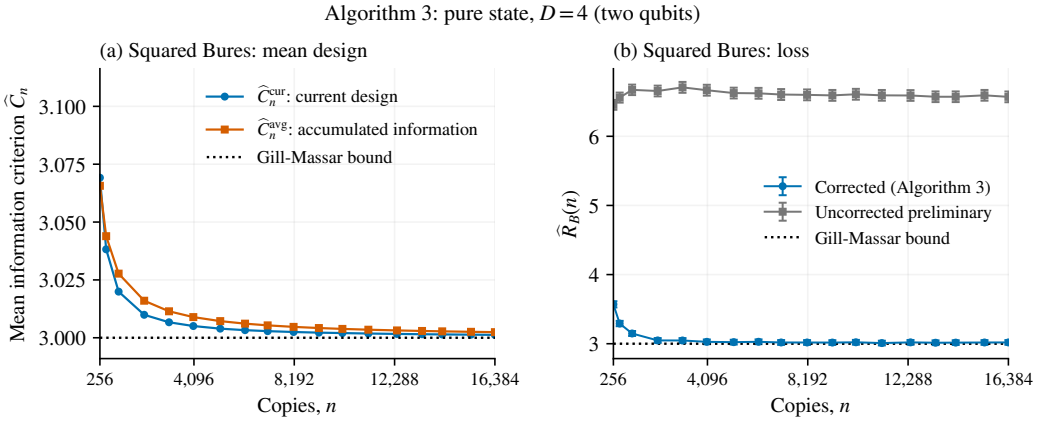


FIG 2. *Squared Bures design and loss for Algorithm 3 at* $D = 4$. *(a) Monte Carlo means* $\widehat{C}_n^{\mathrm{cur}}$ *and* $\widehat{C}_n^{\mathrm{avg}}$ *over 10,000 independent replicates, with shaded* 1.96 *Monte Carlo standard-error bands. (b) Actual scaled loss of the preliminary and corrected outputs on identical records, averaged over 10,000 independent Monte Carlo replicates, with* 1.96 *Monte Carlo standard-error bars. The dotted benchmark is three, as in Corollary 5.3.*

At $n = 16{,}384$, its corrected mean is 3.0188 (MC s.e. 0.0174), compared with the benchmark three; the preliminary mean is 6.5703 (MC s.e. 0.0397).

Together, the figures separate two requirements for attainment: loss-matched measurement design in Figure 1, and an efficient reported estimator for that design in Figure 2. The analytic covariance-inflation example in (4.7) uses a different POVM from Figure 2, which uses Example 4.1; its inflation constant therefore does not predict the plotted preliminary loss.

7. Discussion

We establish strong consistency, Fisher information convergence and asymptotic normality with inverse-information covariance for explicit least-squares procedures under repeated adaptive measurement selection. Adaptive ridge estimation handles interior mixed qubits; a weighted tangent correction gives efficient pure-state estimates in every fixed finite dimension. The analysis allows arbitrary predetermined positive finite batches and accounts for reuse of observations in measurement alignment and estimation. The moment bounds supporting Corollaries 5.2 and 5.3 further establish pointwise expected-loss attainment of the loss-specific Gill–Massar benchmarks.

Supplementary material

Appendices A and B give the pure-state geometry and conditional-likelihood framework. Each theoretical result is followed by its proof location. Supporting probability, geometry and moment estimates, measurement constructions, and reproduction details for Figures 1 and 2 accompany the detailed proofs.

Data and code availability

The Python simulation code and Lean sources are maintained in the GitHub repository `https://github.com/HongruZhao/adaptive-quantum-tomography` (currently private; public release pending). The accompanying versioned archives are `adaptive_quantum_tomography_python_20261018.zip` and `adaptive_quantum_tomography_main_paper_lean_20261018.zip`. They contain, respectively, configurations, replicate arrays and reproduction instructions, and the Lean toolchain, dependencies, build instructions and manuscript-to-declaration correspondence.

Use of generative artificial intelligence

GPT-6 was used for LaTeX preparation, Lean formalization, refinement of proofs to simplify arguments and reduce unnecessary assumptions, and development of numerical code. The authors are responsible for reviewing and editing all assisted content.

Appendix A: Pure-state geometry

We collect the geometric definitions used for the pure-state model $\mathcal{P}_D$, with $D \geq 2$ (Brody and Hughston, 2001). Recall that $d = D - 1$. Coordinates describe nearby projectors and tangent vectors describe their first-order changes (Boumal, 2023, Chapters 3 and 8). Fisher information forms measure the sensitivity of measurement probabilities (Braunstein and Caves, 1994). The definitions below use the projector representation throughout.

Definition A.1 (Pure states, rays and phase equivalence)**.** A pure state is a rank-one orthogonal projector. The projector model is

$$\mathcal{P}_D = \{\psi\psi^* : \psi \in \mathbb{C}^D,\ \|\psi\| = 1\}.$$

For a nonzero vector $\psi \in \mathbb{C}^D$, its ray $[\psi]$ is the one-dimensional complex subspace $\mathbb{C}\psi$. The space of these rays is

$$\mathbb{CP}^{D-1} = (\mathbb{C}^D \setminus \{0\})/\sim, \qquad \psi \sim \widetilde{\psi} \iff \widetilde{\psi} = c\psi \quad (c \in \mathbb{C} \setminus \{0\}).$$

Equivalently, $\mathbb{CP}^{D-1} \cong S^{2D-1}/U(1)$, where $S^{2D-1} = \{\psi \in \mathbb{C}^D : \|\psi\| = 1\}$ and $U(1) = \{c \in \mathbb{C} : |c| = 1\}$ acts freely by $\psi \mapsto c\psi$. Thus unit vectors differing only by a global phase represent the same projector (Ashtekar and Schilling, 1999, Section II B).

The correspondence with the projector manifold is

$$\Phi : \mathbb{CP}^{D-1} \longrightarrow \mathcal{P}_D, \qquad \Phi([\psi]) = \frac{\psi\psi^*}{\psi^*\psi},$$

with inverse sending a projector to its one-dimensional range (Facchi et al., 2010, Section II). With the standard smooth structure on projective space and the submanifold structure on $\mathcal{P}_D \subset \mathrm{Herm}(\mathbb{C}^D)$, this is a diffeomorphism: a smooth bijection with a smooth inverse. In particular, $\cong$ above denotes this diffeomorphism. The pure-state manifold has real dimension $2d$ (Brody and Hughston, 2001). Here and below, Hermitian matrices and their tangent subspaces use real scalars; multiplication by an arbitrary complex scalar need not preserve Hermitian symmetry.

Definition A.2 (Charts and parametrizations of pure states)**.** A chart on $\mathcal{P}_D$ is a pair $(\mathcal{U}, \varphi)$, where $\mathcal{U}$ is open in $\mathcal{P}_D$ and $\varphi : \mathcal{U} \to \mathcal{V} \subset \mathbb{R}^{2d}$ is a homeomorphism onto an open coordinate domain, compatible with the smooth structure. Its inverse $\Gamma(\theta) = \varphi^{-1}(\theta)$ is a parametrization by real coordinates. On overlapping chart domains, the coordinate change $\varphi_2 \circ \varphi_1^{-1}$ is smooth with a smooth inverse. An atlas is a collection of such compatible charts covering $\mathcal{P}_D$ (Lee, 2013, Chapter 1).

Reference coordinates and rotations. The reference parametrization in (4.1) maps $\mathcal{B} \subset \mathbb{R}^{2d}$ onto the open domain

$$\mathcal{U}_{e_0} = \{P \in \mathcal{P}_D : e_0^* P e_0 > 0\}.$$

From coordinates (x, y), we form the unit vector $\psi(x, y)$. This vector determines the ray $[\psi(x, y)]$. The map Φ in Definition A.1 sends this ray to the projector $\Gamma_{\mathrm{ref}}(x, y)$. The convention $e_0^*\psi(x, y) > 0$ fixes the phase of the unit-vector representative. For each $P \in \mathcal{U}_{e_0}$, this representative and the inverse coordinate map are

$$g(P) = \frac{Pe_0}{(e_0^* P e_0)^{1/2}}, \qquad \varphi_{\mathrm{ref}}(P) = (x, y),$$
$$x_r = 2\Re(e_r^* g(P)), \qquad y_r = 2\Im(e_r^* g(P)), \quad r = 1, \ldots, d.$$

Recall that $e_0, \ldots, e_d$ are the fixed orthonormal basis vectors of $\mathbb{C}^D$ introduced in Section 4.1. Here $\Re$ and $\Im$ denote the real and imaginary parts, respectively. Thus $\Gamma_{\mathrm{ref}} = \varphi_{\mathrm{ref}}^{-1}$ as maps, as required in Definition A.2; the square roots have positive arguments on the respective open domains, so both maps are smooth.

For a fixed unitary $R = (h, b_1, \ldots, b_d)$, the rotated parametrization in (4.2) has center hh^*, domain and inverse map

$$\mathcal{U}_h = \{P \in \mathcal{P}_D : h^* P h > 0\}, \qquad \varphi_{h,b}(P) = \varphi_{\mathrm{ref}}(R^* P R).$$

Here $h, b_1, \ldots, b_d$ are fixed choices; only x, y vary within the parametrization. States orthogonal to h lie outside $\mathcal{U}_h$ but remain in the model. Every pure state has nonzero overlap with at least one vector of an orthonormal basis. Choosing the centers from that basis therefore gives an atlas covering $\mathcal{P}_D$. The information matrices in Section 2.4 are coordinate representations in such charts, as in the statistical-geometric formulation of Kass (1989).

Definition A.3 (Tangent directions and horizontal representatives)**.** A tangent vector at $P \in \mathcal{P}_D$ is the derivative $A = \gamma'(0)$ of a smooth curve of pure states with $\gamma(0) = P$. For $P = hh^*$, the tangent space and its horizontal representation are

$$T_P = \{A = A^* : PA + AP = A\} = \{hv^* + vh^* : v \in h^\perp\}, \qquad h^\perp = \{v \in \mathbb{C}^D : h^* v = 0\}.$$

The vector v is the part of a unit-vector velocity orthogonal to h; the omitted direction ih changes only the phase. Horizontal representatives describe tangent directions under the sphere-to-ray map $v \mapsto [v]$ displayed after (4.1) in Section 4.1; the map Φ of Definition A.1 identifies rays with projectors (Ashtekar and Schilling, 1999, Sections II B–C, especially Eq. (2.16)).

Differentiating $\gamma(s)^2 = \gamma(s)$ at $s = 0$ gives $PA + AP = A$. Conversely, let $A = A^*$ satisfy this identity. Then $h^* A h = 0$, so $v = Ah \in h^\perp$ and $A = hv^* + vh^*$. For every $v \in h^\perp$, the curve $s \mapsto (h + sv)(h + sv)^*/(1 + s^2\|v\|^2)$ consists of pure states and has derivative $hv^* + vh^*$ at zero. This proves both descriptions of T_P. Since $v = Ah$ is unique, T_P has real dimension $2d$, twice the complex dimension d of $h^\perp$.

For an inverse chart Γ, the differential at θ maps a coordinate direction $u \in \mathbb{R}^{2d}$ to $\sum_{j=1}^{2d} u_j A_j$, where $A_j = \partial\Gamma(\theta)/\partial\theta_j$. These matrices form a real basis of $T_{\Gamma(\theta)}$. If $\theta = \theta(\eta)$ is a change of coordinates, the chain rule gives

$$\frac{\partial}{\partial \eta_k}\Gamma(\theta(\eta)) = \sum_{j=1}^{2d} A_j J_{jk}, \qquad J_{jk} = \frac{\partial \theta_j}{\partial \eta_k},$$

where A_j is evaluated at $\theta(\eta)$ (Boumal, 2023, Sections 3.4 and 8.8).

Definition A.4 (Information forms and the Fubini–Study metric)**.** A real bilinear form G_P assigns $G_P[A, B] \in \mathbb{R}$ to $A, B \in T_P$, linearly in each argument. A Riemannian metric is a smoothly varying family of symmetric positive definite forms on these tangent spaces. Riemannian structures obtained from the underlying Hilbert space are studied by Provost and Vallée (1980). On pure-state rays we use the Fubini–Study metric with normalization

$$g_{\mathrm{FS},P}[hv^* + vh^*, hw^* + wh^*] = \Re(v^* w), \qquad v, w \in h^\perp,$$

which agrees with the pure-state line element in Stokes et al. (2020, Eqs. (5)–(7)).

Proposition 2.1 gives $\mathcal{H}_P[A, B] = 2\,\mathrm{tr}(AB) = 4g_{\mathrm{FS},P}[A, B]$. The relation between quantum Fisher information and pure-state statistical distance is developed by Braunstein and Caves (1994). Section 2.4 defines the classical information form $\mathcal{J}_{a,P}$ and the coordinate matrices of both information forms (Barndorff-Nielsen and Gill, 2000, Section 2). The classical form can be singular; it is a Riemannian metric only where it is positive definite and smooth. The positive-support convention at zero probabilities does not impose continuity there.

Definition A.5 (Frobenius tangent projection and coordinates)**.** On Hermitian matrices use the real inner product $\langle A, B\rangle_F = \mathrm{tr}(AB)$. The Frobenius-orthogonal projection onto T_P is

$$\Pi_P(A) = PA(\mathrm{I}_D - P) + (\mathrm{I}_D - P)AP, \qquad A = A^*.$$

A Frobenius-orthonormal basis $T_1, \ldots, T_{2d}$ of T_P satisfies $\mathrm{tr}(T_j T_k) = \delta_{jk}$, where δ_{jk} is one if $j = k$ and zero otherwise. It represents $\Delta \in T_P$ by real coefficients $z_j = \mathrm{tr}(T_j \Delta)$.

For $P = hh^*$, the terms $PA(\mathrm{I}_D - P)$ and $(\mathrm{I}_D - P)AP$ map $h^\perp$ into the line $\mathbb{C}h$ and $\mathbb{C}h$ into $h^\perp$, respectively. These are the two off-diagonal blocks of A. The projection keeps their sum and removes PAP and $(\mathrm{I}_D - P)A(\mathrm{I}_D - P)$, which act within these subspaces and are Frobenius-orthogonal to T_P (Boumal, 2023, Section 3.7). Proposition 2.1 gives $H = 2\mathrm{I}_{2d}$ in this basis. These are the tangent coordinates of Algorithm 3, distinct from the amplitude coordinates of Section 4.1, where $H(0) = \mathrm{I}_{2d}$.

Definition A.6 (Normalization retraction for pure states)**.** A retraction on $\mathcal{P}_D$ is a smooth map taking (P, Δ), with $\Delta \in T_P$, to a pure state $R_P(\Delta)$, such that

$$R_P(0) = P, \qquad \frac{\mathrm{d}}{\mathrm{d}s} R_P(s\Delta)\Big|_{s=0} = \Delta, \qquad \Delta \in T_P.$$

The derivative is taken with respect to the real scalar s, with P and Δ fixed. Equivalently, the differential of $\Delta \mapsto R_P(\Delta)$ at zero is the identity on T_P (Boumal, 2023, Section 3.6). In the notation $\mathrm{D}R_P(0)$, D denotes differentiation. The normalization used in Algorithm 3 is

$$R_P(\Delta) = \frac{(\mathrm{I}_D + \Delta)P(\mathrm{I}_D + \Delta)}{1 + \|\Delta h\|^2}, \qquad P = hh^*.$$

Since $h^*\Delta h = 0$, this is the projector of the normalized vector $h + \Delta h$. It is independent of the phase of h and satisfies $R_P(\Delta) = P + \Delta + O(\|\Delta\|_F^2)$ as $\Delta \to 0$. The supplement derives this exact formula and the quadratic remainder in Section S4, equation (S4.1). The retraction therefore returns the linear tangent correction to the pure-state model while preserving its first-order change.

Appendix B: Conditional likelihood for the adaptive Gill–Massar bound

This appendix gives the conditional-likelihood assumptions and score conventions used in Theorem 5.1. Let θ_0 lie in an open coordinate neighborhood in $\mathbb{R}^p$. Use the parameter-independent policies of Section 5; for a fixed history h, their conditional probabilities are

$$p_{t,\theta}(o \mid \mathsf{h}) = \mathrm{tr}\{\rho(\theta) M_{t,o}(\mathsf{h})\}.$$

At each fixed sample size, represent the record labels by finite sets independent of θ, padding with zero-probability labels when needed. Assume all first partial derivatives of these probabilities exist at θ_0 for every history and outcome. For fixed h, only $\rho(\theta)$ is differentiated; the measurement effects are held fixed. The distribution of the history may still depend on θ. The joint likelihood is

$$p_\theta^{(n)}(o_1, \ldots, o_n) = \prod_{t=1}^{n} p_{t,\theta}(o_t \mid o_1, \ldots, o_{t-1}). \tag{B.1}$$

These conditions permit differentiation of finite sums term by term.

Scores and conditional information. If necessary, extend each conditional law outside this neighborhood by its value at the truth; this leaves every local statement unchanged. At a zero-probability outcome each derivative is zero, since a differentiable nonnegative function has a local minimum there. Define its score to be zero; otherwise set $s_t(o, \mathsf{h}) = \nabla_\theta p_{t,\theta_0}(o \mid \mathsf{h})/p_{t,\theta_0}(o \mid \mathsf{h})$. The conditional Fisher information matrix is

$$\mathcal{I}_t(\mathsf{h}) = \sum_o p_{t,\theta_0}(o \mid \mathsf{h}) s_t(o, \mathsf{h}) s_t(o, \mathsf{h})^{\mathsf{T}}.$$

The score calculation uses first derivatives of the conditional probabilities at θ_0. It assumes neither a likelihood expansion nor continuity of Fisher information at zero-probability outcomes.

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